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IWASAKI

(54) **VEHICLE ELECTRONIC COMPONENT HOLDER SET AND VEHICLE ELECTRONIC COMPONENT HOLDER**

(71) Applicant: **SUMIDA CORPORATION**, Tokyo (JP)

(72) Inventor: **Noriaki IWASAKI**, Natori City (JP)

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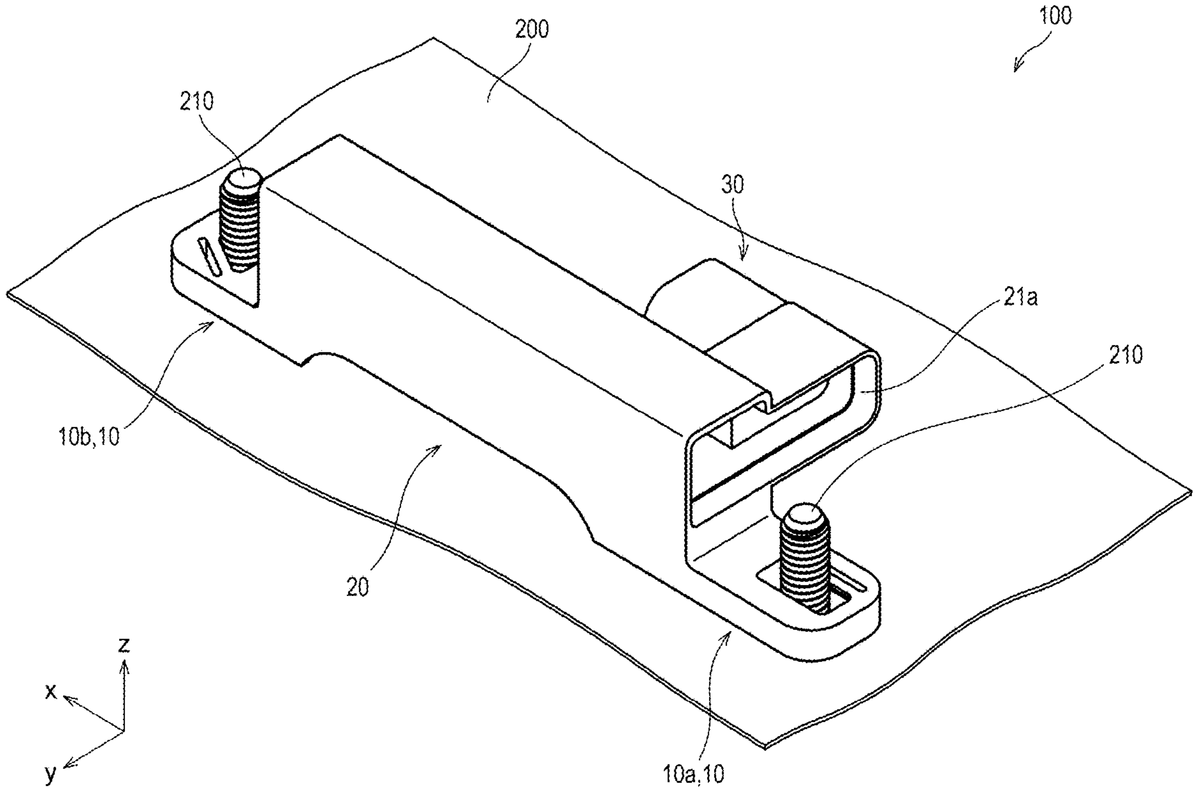
CPC ***B60R 16/0239*** (2013.01)

(57) **ABSTRACT**

A vehicle electronic component holder set includes a shaft member having a circular cross section, and a vehicle electronic component holder. The vehicle electronic component holder includes a base portion provided with an attachment hole through which the shaft member is inserted. The base portion includes a protrusion protruding from a peripheral wall defining the attachment hole toward an inside of the attachment hole. As viewed from the depth direction of the attachment hole, a distance between the protrusion and a center of the attachment hole is smaller than a radius of the shaft member. When the shaft member is inserted into the attachment hole, the protrusion comes into pressure contact with a side surface of the shaft member.



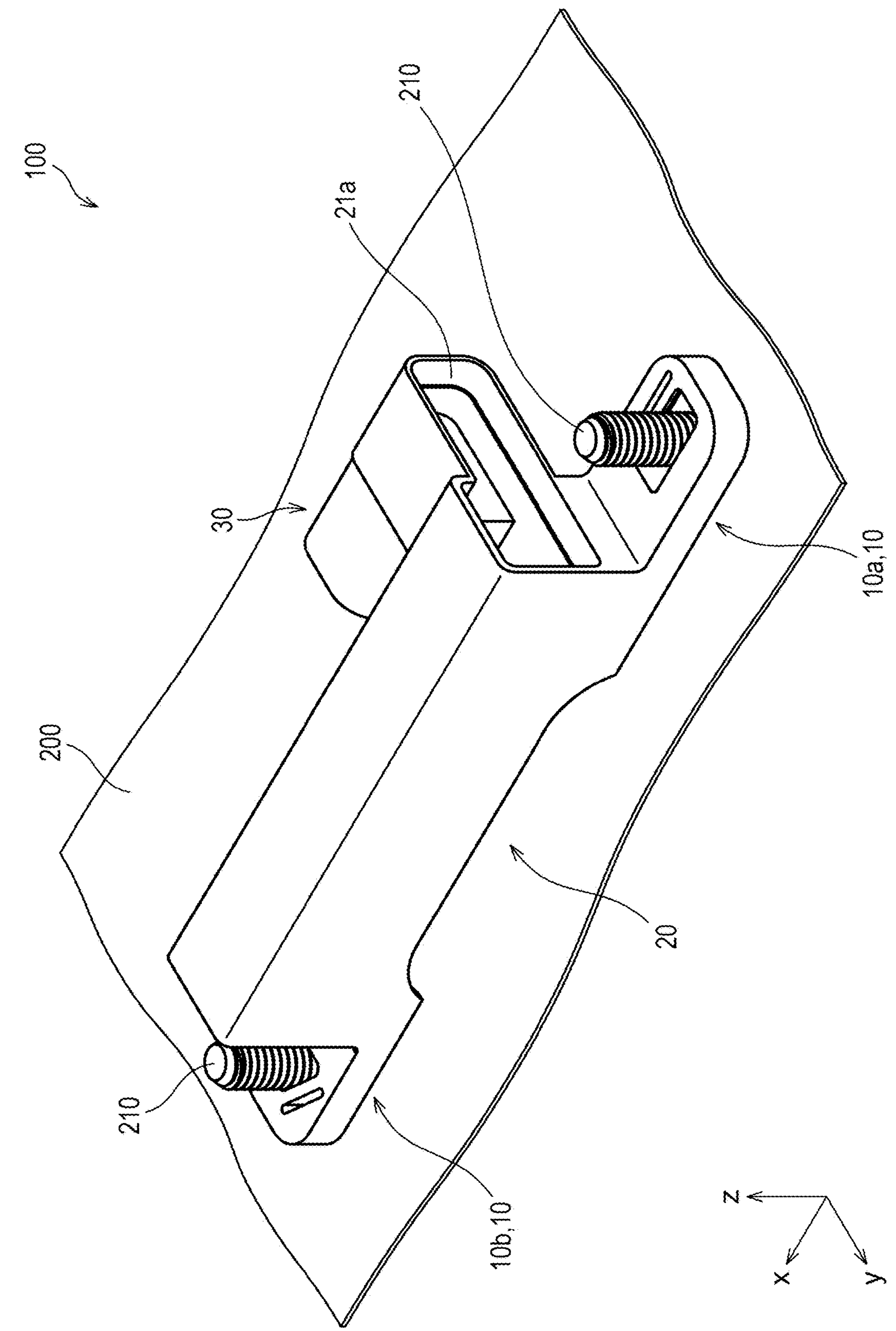
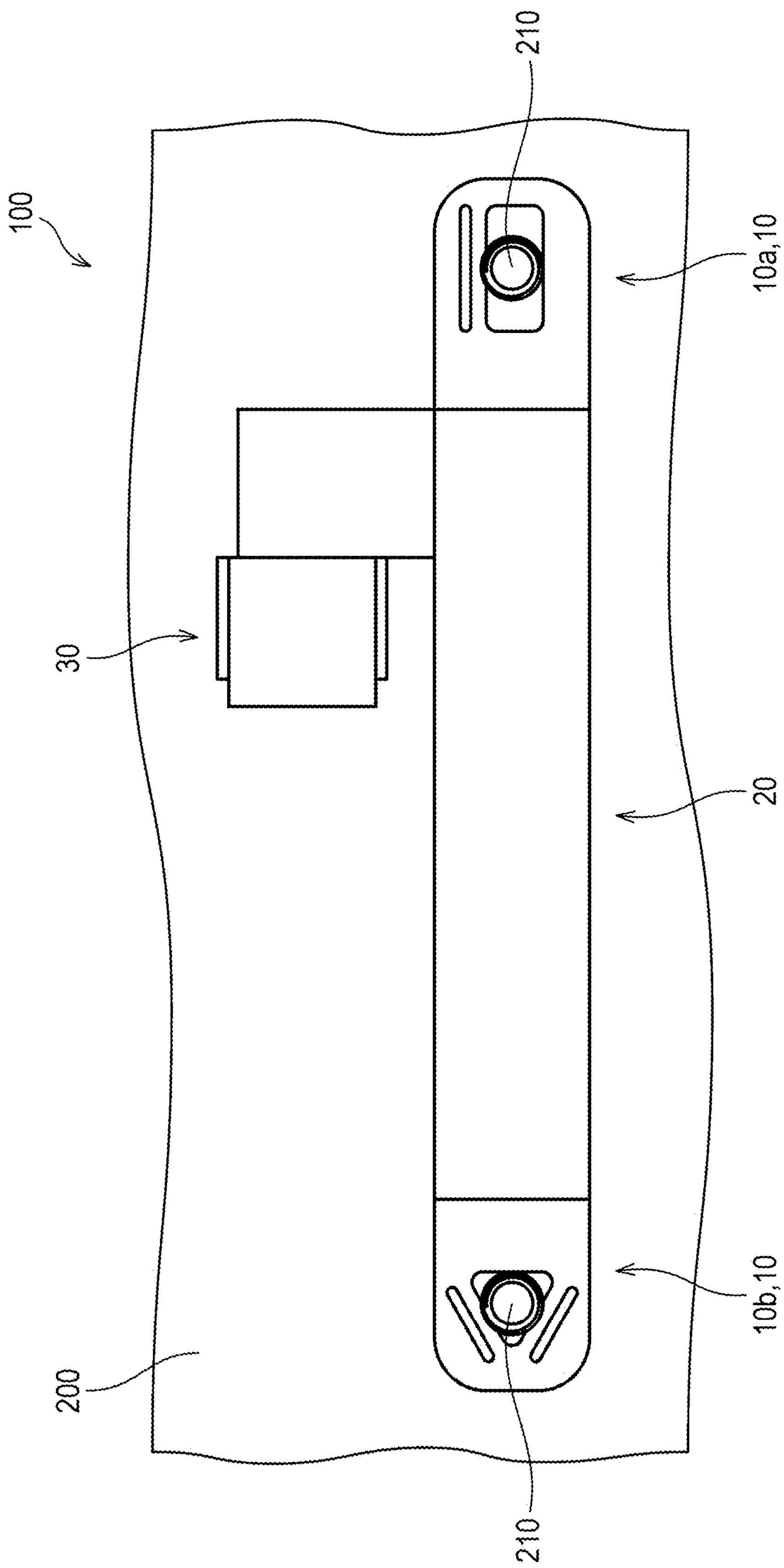


FIG.1

FIG.2



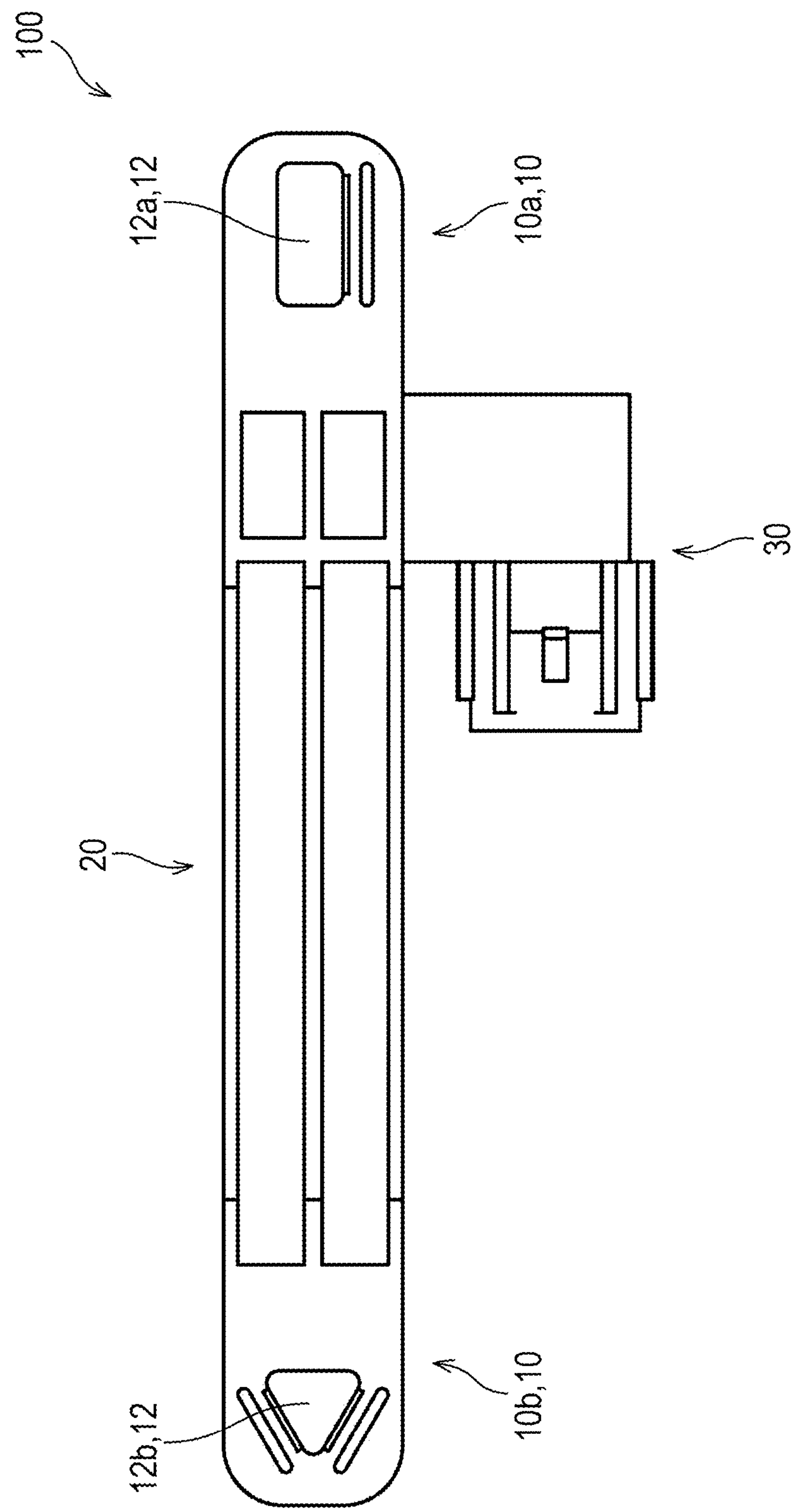


FIG.3

FIG.4

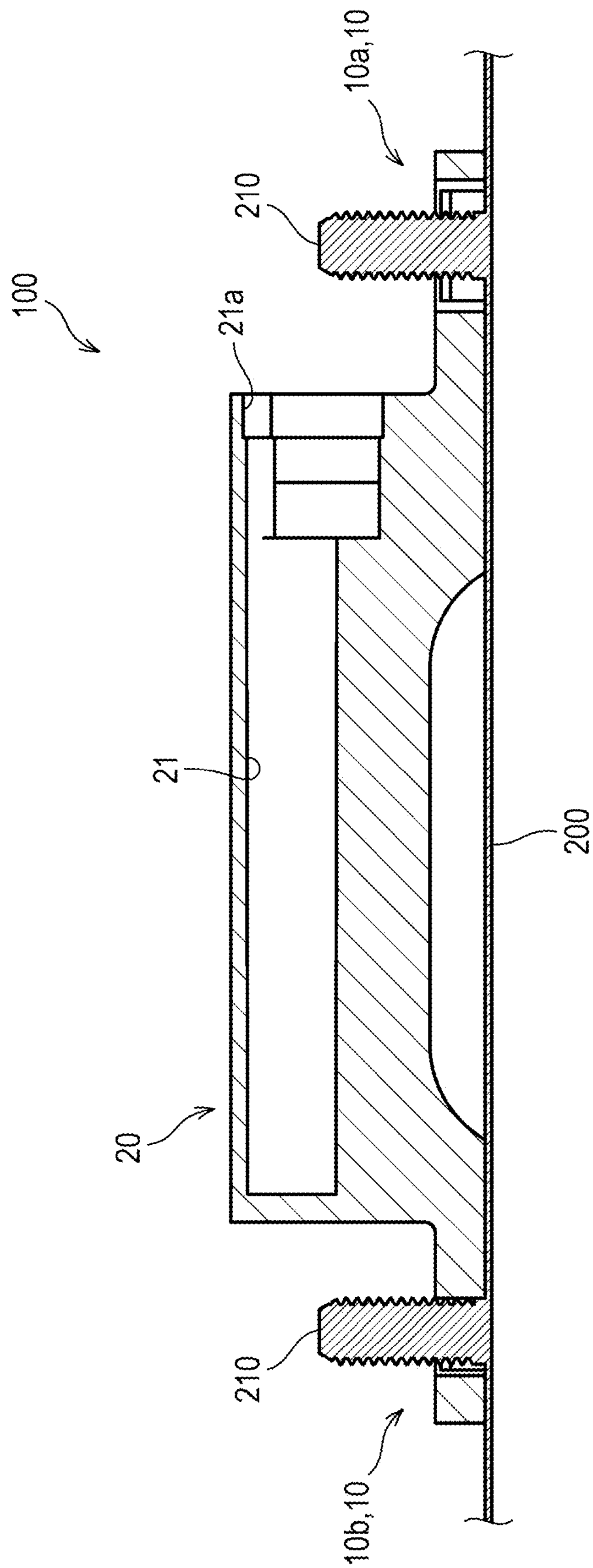


FIG.5A

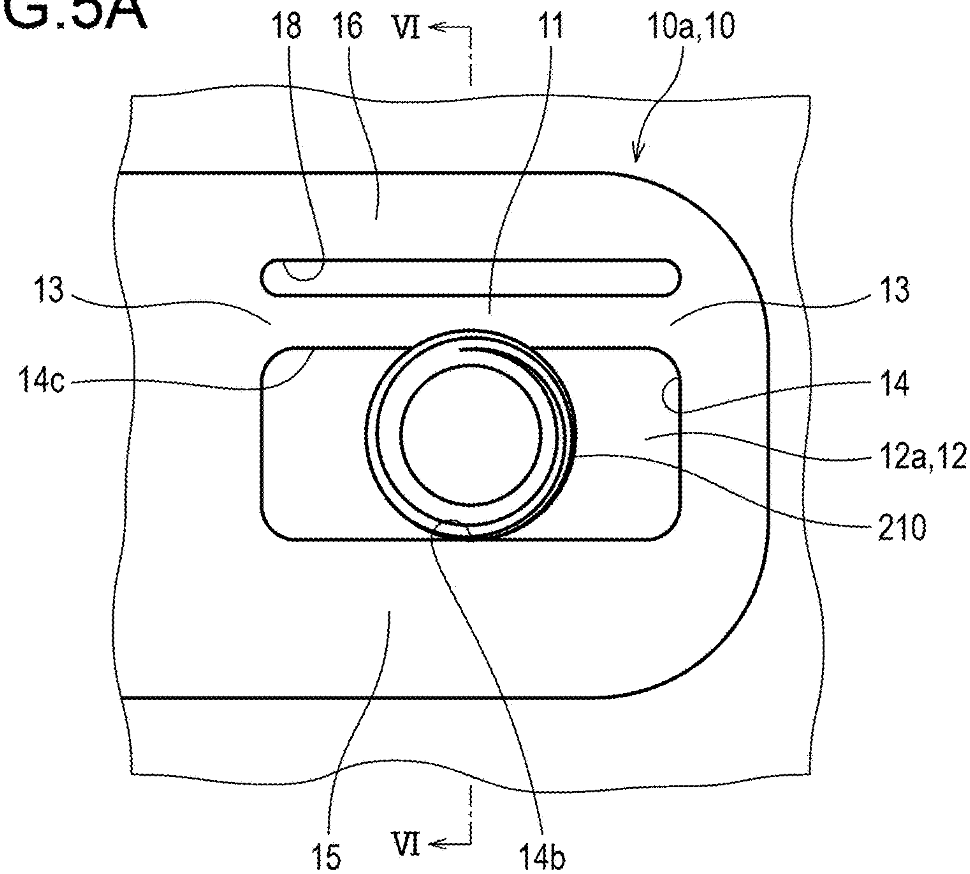

18
16
VI
11
10a,10
13
13
14c
14
12a,12
210
15
VI
14b

FIG.5B

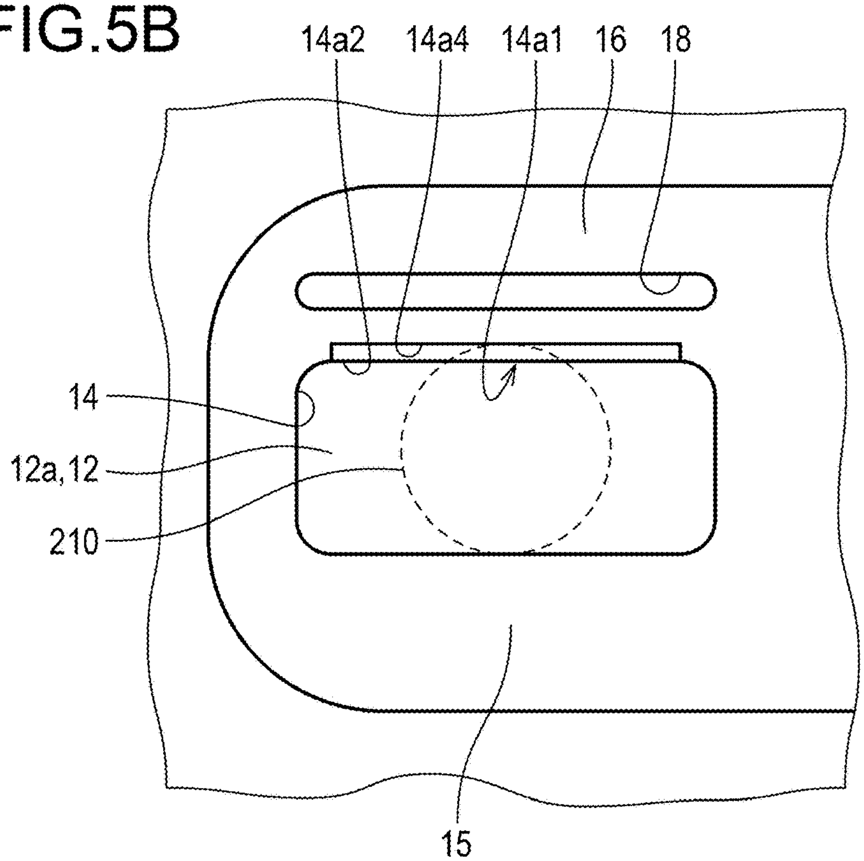

14a2
14a4
14a1
16
18
14
12a,12
210
15

FIG.6A

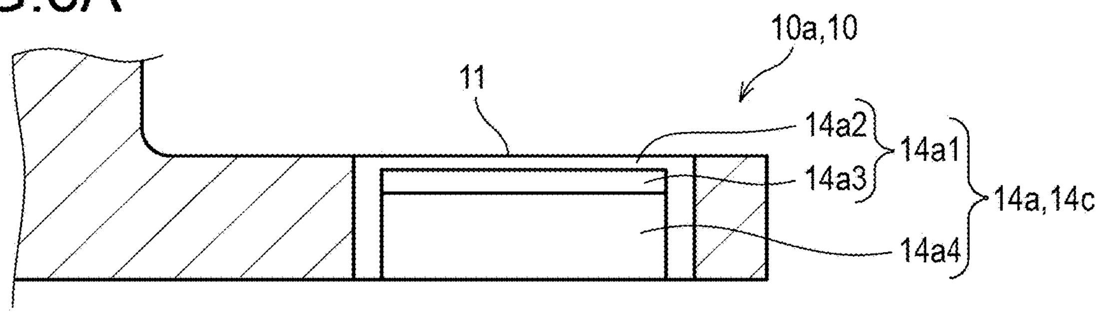


FIG.6B

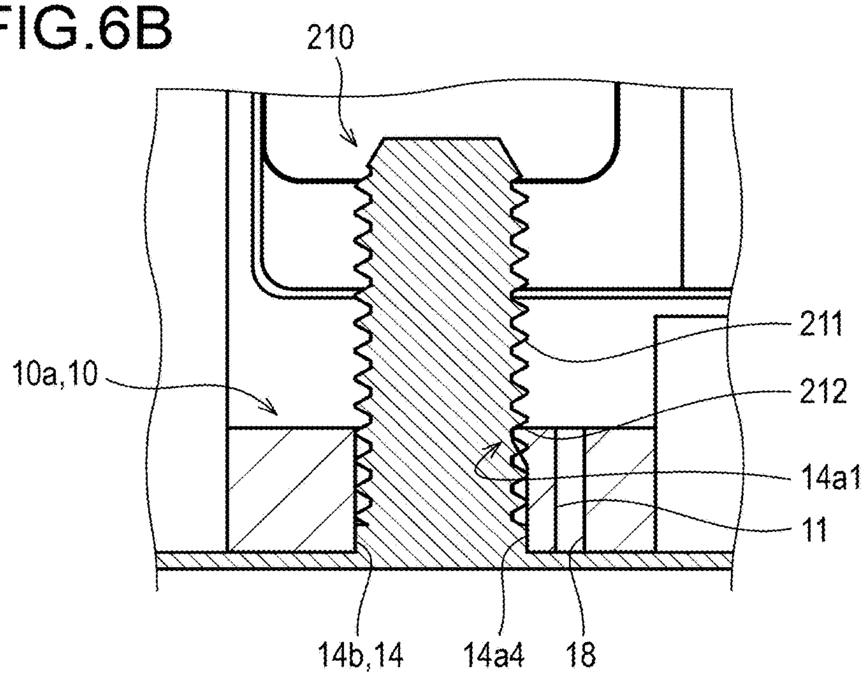


FIG.6C

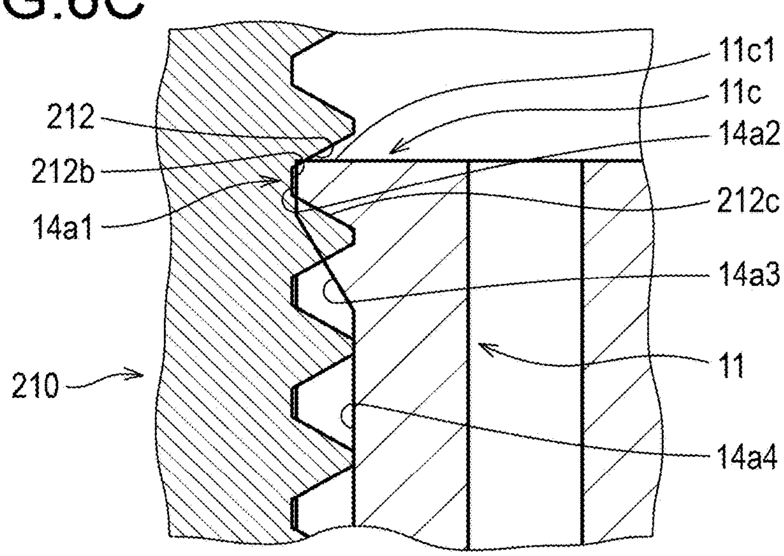
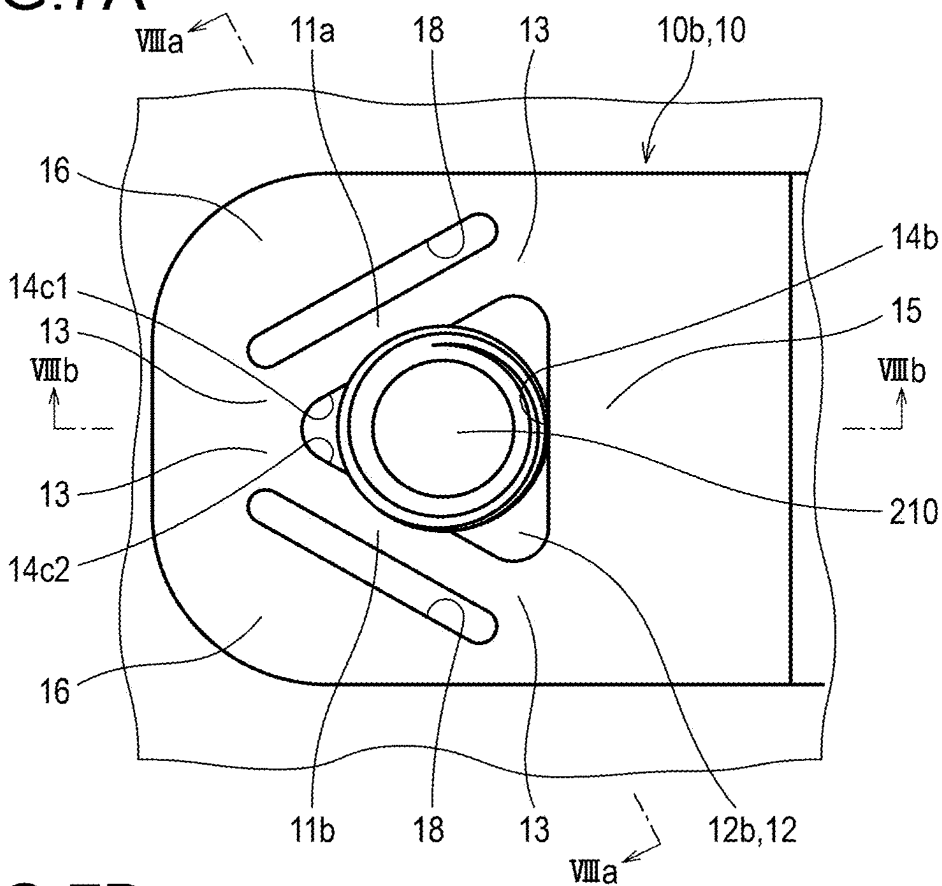
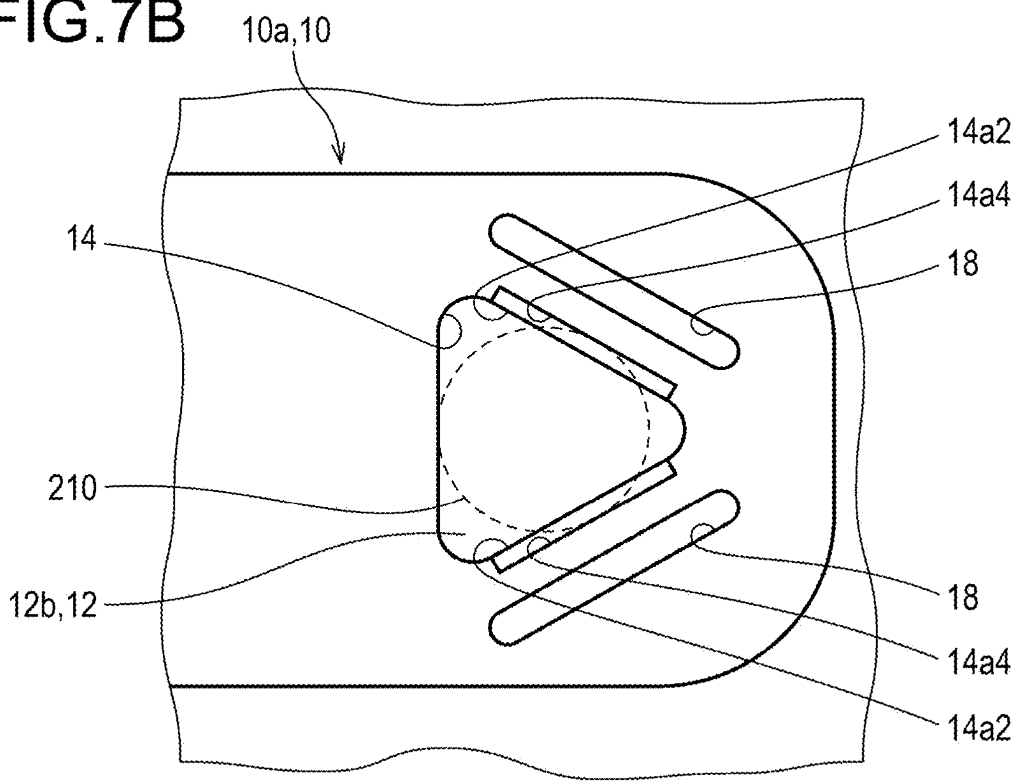


FIG.7A



VIIIa
11a
18
13
10b,10
16
14b
14c1
15
13
VIIIb
VIIIb
13
210
14c2
16
11b
18
13
12b,12
VIIIa

FIG.7B



10a,10
14a2
14a4
14
18
210
18
12b,12
14a4
14a2

FIG.8A

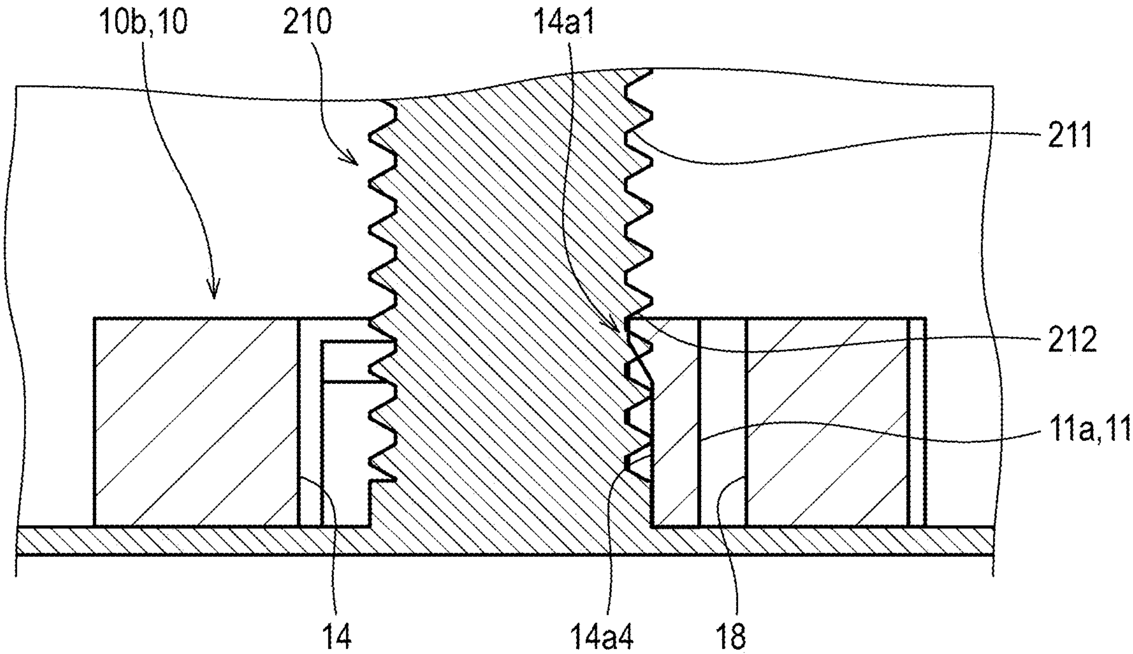
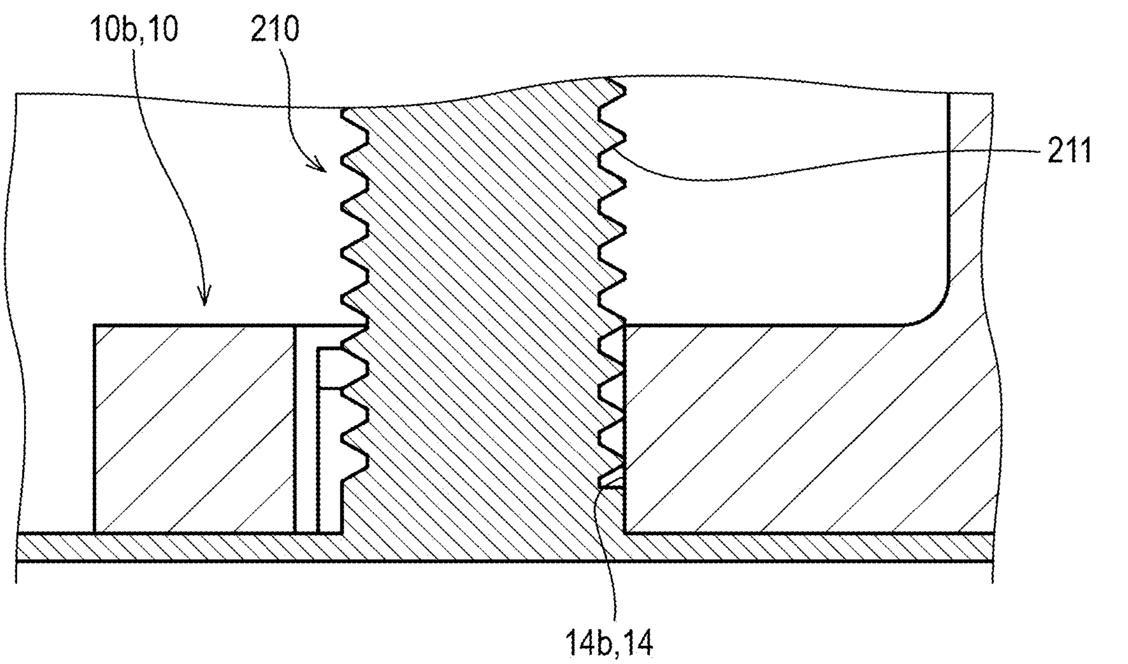


FIG.8B



VEHICLE ELECTRONIC COMPONENT HOLDER SET AND VEHICLE ELECTRONIC COMPONENT HOLDER

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to International Patent Application No. PCT/JP2022/046639, filed on Dec. 19, 2022, which is expressly incorporated herein by reference in its entirety.

BACKGROUND

Technical Field

[0002] The present invention relates to a vehicle electronic component holder set and a vehicle electronic component holder.

Related Art

[0003] A holder may be used to attach and hold a coil component to another member.

[0004] Regarding this type of technology, JP 2016-100610 A discloses an antenna device (**10**) that is attached to another member such as a bumper (**31***a*) of a vehicle body (**30***a*) using a case (**3**) having a case main body (**3***b*) in which an antenna coil (**1**) is accommodated and a flange portion (**3***f*). Specifically, as shown in FIG. **2** of the same document, the flange portion (**3***f*) has a through hole (**3***fb*), and the antenna device (**10**) is attached to another member by inserting an attachment screw into the through hole (**3***fb*) and fastening the attachment screw.

[0005] When the flange portion (**3***f*) provided with the through hole (**3***fb*) is attached to an installation surface of another member, a shaft portion of an attachment screw is inserted into the through hole (**3***fb*) and the screw hole of the installation surface, and the flange portion (**3***f*) is sandwiched and held between a screw head having a diameter larger than that of the through hole (**3***fb*) and the installation surface, whereby the antenna device (**10**) is held by the another member. In general, the through hole (**3***fb*) is formed so that the diameter of the through hole (**3***fb*) is larger than the outer diameter of the shaft portion of the attachment screw. Thus, only by inserting the mounting screw into the through hole (**3***fb*) and the screw hole, the relative position of the antenna device with respect to the attachment screw and other members is not fixed. That is, it is necessary to fix the antenna device (**10**) at a desired mounting position on another member by the installer himself or herself or some sort of tool until the mounting screw is completely tightened. As described above, when the antenna device of JP 2016-100610 A is attached to another member, it takes time and effort to hold the antenna device at a desired attachment position of the another member.

[0006] The present invention has been made in view of the above-described problems, and an object of the present invention is to provide an electronic component holder that is easily attached.

SUMMARY

[0007] A vehicle electronic component holder set according to the present invention is a vehicle electronic component holder set including: a shaft member having a circular cross section; and a vehicle electronic component holder, wherein the vehicle electronic component holder includes a base portion provided with an attachment hole through which the shaft member is inserted, the base portion includes a protrusion protruding from a peripheral wall defining the attachment hole toward an inside of the attachment hole, a distance between the protrusion and a center of the attachment hole is smaller than a radius of the shaft member as viewed from a depth direction of the attachment hole, and when the shaft member is inserted into the attachment hole, the protrusion comes into pressure contact with a side surface of the shaft member.

[0008] A vehicle electronic component holder according to the present invention is a vehicle electronic component holder used together with a shaft member having a circular cross section, the vehicle electronic component holder including: a base portion provided with an attachment hole through which the shaft member is inserted, wherein the base portion includes a protrusion protruding from a peripheral wall defining the attachment hole toward an inside of the attachment hole.

[0009] Since the distance between the protrusion and the center of the attachment hole as viewed from the depth direction of the attachment hole is smaller than the radius of the shaft member, when the shaft member is inserted into the attachment hole, the peripheral wall including the protrusion is pushed and expanded outward by the shaft member, and the shaft member is gripped by the protrusion and its facing portion. Therefore, the electronic component holder is fixed to the shaft member only by inserting the shaft member into the attachment hole.

Effect of the Invention

[0010] According to a vehicle electronic component holder set of the present invention, when a shaft member is inserted into an attachment hole, a relative position between the shaft member and the electronic component holder is fixed, so that the electronic component can be easily installed on another member.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The foregoing and other objects, features, and advantages will become more apparent from the following preferred embodiments and accompanying drawings.

[0012] FIG. **1** is a perspective view showing an example of a vehicle electronic component holder set in an attached state according to a first embodiment of the present invention.

[0013] FIG. **2** is a top view of the vehicle electronic component holder set in the attached state according to the first embodiment.

[0014] FIG. **3** is a bottom view of the vehicle electronic component holder according to the first embodiment.

[0015] FIG. **4** is a longitudinal sectional view along an x-axis direction of the vehicle electronic component holder set in the attached state according to the first embodiment.

[0016] FIG. **5**A is a top view showing a base portion of the vehicle electronic component holder set in an attached state according to the first embodiment.

[0017] FIG. **5**B is a bottom view showing the base portion of the vehicle electronic component holder in the attached state according to the first embodiment. In FIG. **5**B, an installation surface is not shown, and only an outer edge of a shaft member is indicated by a dotted line.

[0018] FIG. **6**A is a longitudinal cross-sectional view along the x-axis direction of the base portion of the vehicle electronic component holder in the attached state according to the first embodiment.

[0019] FIG. **6**B is a cross-sectional view of a cross section taken along a one-dot chain line shown in FIG. **5**A as viewed in a direction of arrows VI-VI.

[0020] FIG. **6**C is an enlarged view of a protrusion and the vicinity thereof in the drawing of FIG. **6**B.

[0021] FIG. **7**A is a top view showing a base portion different from that in FIG. **5**A of the vehicle electronic component holder set in the attached state according to the first embodiment.

[0022] FIG. **7**B is a bottom view showing a base portion different from that in FIG. **5**B of the vehicle electronic component holder set in the attached state according to the first embodiment. In FIG. **7**B, an installation surface is not shown, and only an outer edge of a shaft member is indicated by a dotted line.

[0023] FIG. **8**A is a cross-sectional view of the vehicle electronic component holder in the attached state according to the first embodiment as viewed in a direction of arrows VIIIa-VIIIa along a one-dot chain line shown in FIG. **7**A.

[0024] FIG. **8**B is a cross-sectional view of the vehicle electronic component holder in the attached state according to the first embodiment as viewed in a direction of arrows VIIIb-VIIIb along a one-dot chain line shown in FIG. **7**A.

DETAILED DESCRIPTION

[0025] The various constituent elements of the coil component of the present invention do not need to be independent from each other, and it is allowable that a plurality of constituent elements is formed as one member, one constituent element is formed of a plurality of members, one constituent element is a part of another constituent element, a part of one constituent element overlaps with a part of another constituent element, and the like.

[0026] Hereinafter, embodiments of the present invention will be described with reference to the drawings. In the drawings, corresponding components are denoted by the same reference numerals, and redundant description will not be repeated as appropriate.

[0027] In the present embodiment, an electronic component holder set will be described by defining x, y, and z coordinates shown in FIG. **1**. A z-axis direction coincides with an axial direction of a shaft member and a depth direction of an attachment hole. In the present embodiment, a side on which another member (also referred to as an attachment destination member) to which an electronic component holder is to be attached is referred to as a lower side, a side opposite to the lower side is referred to as an upper side, and a direction (x-axis direction or y-axis direction) orthogonal to a vertical direction (z-axis direction) is referred to as a lateral direction or a side direction. However, this is defined for convenience in order to simply describe the relative relationship of the components, and does not limit the direction at the time of manufacturing or using the product for carrying out the present invention. Further, a center side of the electronic component holder is referred to as an inner side, and a side opposite to the inner side is referred to as an outer side.

[0028] In addition, the plane referred to in the present invention means a shape physically formed with a plane as a target, and it is needless to say that it is not necessarily a geometrically perfect plane.

First Embodiment

(Electronic Component Holder Set)

[0029] FIG. **1** is a perspective view showing an example of a vehicle electronic component holder set according to a first embodiment of the present invention.

[0030] First, an outline of a vehicle electronic component holder set according to the present embodiment will be described.

[0031] The vehicle electronic component holder set (hereinafter, also simply referred to as an electronic component holder set) includes a shaft member **210** having a circular cross section, and a vehicle electronic component holder (hereinafter, also simply referred to as the electronic component holder **100**). The electronic component holder **100** has a base portion **10** provided with an attachment hole **12** (see FIG. **5**A) through which the shaft member **210** is inserted. The base portion **10** has a protrusion **14***al* (see FIG. **6**B) protruding from a peripheral wall **14** (see FIG. **5**A) defining the attachment hole **12** toward the inside of the attachment hole **12**. When viewed from the depth direction of the attachment hole **12**, a distance between the protrusion **14***a***1** and a center of the attachment hole is smaller than the radius of the shaft member **210**. When the shaft member **210** is inserted into the attachment hole **12**, the protrusion **14***a***1** comes into pressure contact with a side surface of the shaft member **210**.

[0032] With the configuration of the electronic component holder **100** as described above, when the shaft member **210** is inserted into the attachment hole **12** as described later, the peripheral wall **14** including the protrusion **14***a***1** is pushed and expanded by the shaft member **210**. Since the peripheral wall **14** pushed and expanded tends to deform toward a center of the attachment hole **12** by its own elastic restoring force, the protrusion **14***al* comes into pressure contact with the side surface of the shaft member **210**. That is, since the shaft member **210** is gripped by the protrusion **14***al* and one or more facing portions **14***b* to be described later, the base portion **10** is fixed to the shaft member **210**. Thus, the electronic component holder **100** can be fixed to the periphery of the shaft member **210** in a desired positional relationship only by disposing the base portion **10** so that the shaft member **210** is inserted into the attachment hole **12**.

[0033] Next, the vehicle electronic component holder set of the present embodiment will be described in detail.

[0034] The vehicle electronic component holder set includes a vehicle electronic component holder and one or a plurality of other members. In addition, the vehicle electronic component holder is a member for attaching and holding an electronic component at a desired position of another member (hereinafter, also referred to as an attachment destination member) in the vehicle. Examples of a place where the electronic component is mounted include a wall portion in the vehicle, a ceiling, a vehicle body such as above a floor or below a floor, or a part in a console box or a trunk room. The electronic component holder **100** may be attached to a desired position in the vehicle in any orientation. For example, in the attached state, any of the x-axis direction, the y-axis direction, and the z-axis direction may be the direction in which gravity is applied.

[0035] Here, the electronic component is a component constituting a device mounted on a vehicle, and a component having a coil is exemplified. Examples of the coil component include an antenna.

[0036] The electronic component holder **100** may hold the electronic component on the attachment destination member by itself, or may hold the electronic component on the attachment destination member by being used together with another member. For example, as described later, a case portion **20** to which an electronic component is attached may be provided as a separate member from the flange portion **10**, and the base portion **10** may connect the case portion **20** and the attachment destination member to each other, so that the base portion **10** and the case portion **20** may function as the electronic component holder **100**.

[0037] The electronic component holder **100** of the present embodiment is integrally formed of resin. As shown in FIGS. **2** and **3**, the electronic component holder **100** has an elongated outer shape whose longitudinal direction is the x-axis direction as a whole. In particular, the case portion **20** described later is a member elongated in the x-axis direction. Although an installation surface **200** shown in FIG. **1** is a flat surface and represents a part of a surface of the attachment destination member, the installation surface **200** may be a surface having irregularities instead of a flat surface, or may be a curved surface.

[0038] The base portion **10** in the electronic component holder **100** is a portion connected to the shaft member **210**. Specifically, the base portion **10** is a member including a portion where the attachment hole **12** to be described later is formed and the vicinity thereof.

[0039] In the present embodiment, the electronic component holder **100** has a case portion **20** described later in detail, and as shown in FIG. **2**, two plate-like members (referred to as flange portions) are provided so as to protrude from the case portion **20** toward both sides in the x-axis direction in a flange shape. In other words, the electronic component holder **100** has two flange portions at both ends in the longitudinal direction of the electronic component holder **100**. As shown in FIG. **3**, the attachment hole **12** is formed substantially at a center of each of the flange portions. The base portion **10** in the present embodiment is a flange portion (hereinafter referred to as the flange portion **10**) provided with the attachment hole **12**. Unlike the present embodiment in which the attachment hole **12** is provided at a center of the flange portion **10**, when the attachment hole **12** is formed on a proximal end side (the case portion **20** side) of the flange portion **10**, a part of the case portion **20** (a part of the flange portion **10** side) may also be referred to as the base portion **10** in addition to the flange portion **10**.

[0040] As shown in FIG. **4**, in the present embodiment, the base portion **10** is provided below a center (on the installation surface **200** side) in the thickness direction (z-axis direction) of the electronic component holder **100**. More specifically, the base portion **10** is disposed on the lowermost side of the electronic component holder **100** in the z-axis direction, and a lower surface of the base portion **10** is the lowermost surface of the electronic component holder **100**. The base portion **10** may be provided in the middle of the electronic component holder **100** in the z-axis direction. That is, a lower surface of the case portion **20** may be in contact with the installation surface **200**, and the flange portion **10** may be separated from the installation surface **200**. Further, a spacer member (not shown in the drawings) may be interposed between the electronic component holder **100** and the installation surface **200**.

[0041] As shown in FIGS. **2** and **4**, the flange portion **10** has a flat shape in which a dimension in the z-axis direction is sufficiently smaller than dimensions in the x-axis direction and the y-axis direction and which extends in the lateral direction. When viewed in the z-axis direction, the flange portion **10** has a rectangular shape that is horizontally long in the x-axis direction. The surface of the flange portion **10** facing the z-axis direction is referred to as a main surface. Unlike the present embodiment, even when the base portion **10** has a large dimension in the z-axis direction and the base portion **10** is not a plate-like member, a surface of the base portion **10** facing the z-axis direction is referred to as a main surface.

[0042] The base portion **10** is not limited to have a flange shape protruding from the case portion **20**. For example, the base portion **10** may be included in an envelope volume of the case portion **20** without protruding from the case portion **20**, or may have a shape in which the base portion **10** and the case portion **20** cannot be clearly separated at a glance. Further, the base portion **10** is not limited to the plate shape, and may have a thickness (dimension in the z-axis direction) larger than the dimension in the lateral direction.

[0043] Further, as described later in the modification, the electronic component holder **100** is not limited to have the two base portions **10**, and may have one or three or more base portions **10**.

[0044] The attachment hole **12** is a hole through which the shaft member **210** is inserted. In the present embodiment, as shown in FIG. **3**, the attachment hole **12** is formed as, but is not limited to, a through hole penetrating the flange portion **10** in the z-axis direction. When the base portion **10** is sufficiently thick, the attachment hole **12** may be a recess provided in the base portion **10** and having a bottom surface. In this case, the shaft member **210** is inserted from the opening side of the attachment hole **12**.

[0045] In the present embodiment, as described above, each of the two flange portions **10** (first flange portion **10*a*** and second flange portion **10*b***) is provided with one attachment hole **12** (first attachment hole **12*a*** or second attachment hole **12*b***). One or a plurality of attachment holes **12** may be provided for one flange portion **10**.

[0046] In the present embodiment, the first attachment hole **12*a*** is a horizontally long rectangle in the x-axis direction as viewed in the z-axis direction, and the second attachment hole **12*b*** is an equilateral triangle as viewed in the z-axis direction. The shape of the attachment hole **12** as viewed from the z-axis direction may be any shape, and examples thereof include a circular shape, an elliptical shape, and a polygonal shape.

[0047] Hereinafter, first, the attachment hole **12** and the flange portion **10** will be described while exemplifying the first attachment hole **12*a*** and the first flange portion **10*a***. Features of the attachment hole **12** and the flange portion **10** described below also apply to the second attachment hole **12*b*** and the second flange portion **10*b***. Specific features of the second attachment hole **12*b*** and the second flange portion **10*b*** will be described later.

[0048] As shown in FIGS. **6**A and **6**B, the attachment hole **12** is defined by a peripheral wall **14** having a dimension in a depth direction (z-axis direction) of the attachment hole **12**.

[0049] In the present embodiment, the peripheral wall **14** of the attachment hole **12** is formed so that a part of the surface is recessed and the remaining portion protrudes. Specifically, as shown in FIGS. **6**A and **6**B, in a protrusion forming region **14***a* to be described later, a part of a wall surface (a recessed surface **14***a***4** to be described later) on the lower end side is recessed in the −y direction (the depth side of the paper surface in FIG. **6**A) with respect to a part (the protrusion **14***a***1**) on the upper end side. The recessed surface **14***a***4** is also a part of the peripheral wall **14**.

[0050] The protrusion **14***a***1** is a portion protruding toward the inside of the attachment hole **12** from a part of the peripheral wall **14**.

[0051] Hereinafter, as shown in FIG. **6**A, a part of the peripheral wall **14** on which the protrusion **14***al* is formed is referred to as a protrusion forming region **14***a*. The protrusion forming region **14***a* is a part of the peripheral wall **14** overlapping the protrusion **14***a***1** in the depth direction (z direction) of the attachment hole **12**. In the present embodiment, as shown in FIG. **5**A, the shape of the attachment hole **12** as viewed in the z-axis direction is a polygon, and the peripheral wall **14** has a flat portion constituting one side of the polygon. As shown in FIG. **6**A, in the present embodiment, the entire one flat portion is a protrusion forming region **14***a*. In the attachment hole **12** of the present embodiment, a protrusion **14***a***1** is formed in a part (uppermost end portion) of the protrusion forming region **14***a* on the upper side. Specifically, as shown in FIGS. **5**B and **6**B, a part of the protrusion forming region **14***a* on the lower side (recessed surface **14***a***4**) is formed to be recessed from a part of the peripheral wall **14** connected to the protrusion forming region **14***a* in the circumferential direction and a top surface **14***a***2** to be described later. In addition, the protrusion **14***a***1** (particularly, the top surface **14***a***2**) protrudes toward the inside of the attachment hole **12** more than the recessed surface **14***a***4**. Here, the protrusion **14***a***1** protruding toward the inside of the attachment hole **12** means that the protrusion protrudes toward the side approaching the center of the attachment hole **12** in plan view. However, the protrusion **14***a***1** does not necessarily protrude straight toward the center of the attachment hole **12**. In the present embodiment, the protrusion **14***al* protrudes toward the side approaching the center of the attachment hole **12** in the direction perpendicular to the surface of the protrusion forming region **14***a*. In other words, the protrusion **14***a***1** protrudes so as to stand in a direction perpendicular to the surface of the protrusion forming region **14***a*.

[0052] As described above, the peripheral wall **14** is provided with an uneven shape, but as shown in FIG. **5**A, the outer edge of the attachment hole **12** in a plan view is configured so that a curve and a straight line are smoothly connected. In other words, as shown in FIG. **5**B, the tip of the protrusion **14***al* (a top surface **14***a***2** to be described later) is smoothly connected to a part of the peripheral wall **14** continuous with the protrusion forming region **14***a* in the circumferential direction (here, in the peripheral wall **14**, a curved portion defining a corner of the attachment hole **12** having a polygonal shape is formed) without a step. That is, both ends of the top surface **14***a***2** are smoothly connected to the curved portion without a step.

[0053] In the first attachment hole **12***a*, two flat portions (facing portion **14***b*, attachment hole facing surface **14***c*) are arranged substantially parallel to each other, and one flat portion (attachment hole facing surface **14***c*) entirely constitutes the protrusion forming region **14***a*. As a result, the side surface of the shaft member **210** and the protrusion **14***a***1** come into pressure contact with each other in substantially the same mode regardless of where the shaft member **210** is disposed in a center or an end of the first attachment hole **12***a* in an extending direction (x-axis direction) of the flat portion (attachment hole facing surface **14***c*).

[0054] Instead of the present embodiment, only a part of the flat portion (attachment hole facing surface **14***c*) may be the protrusion forming region **14***a*. For example, a part of one end side of the flat portion in an extending direction of a leaf spring portion **11** described later or an intermediate portion of the flat portion in the extending direction of the leaf spring portion **11** may be the protrusion forming region **14***a*. When a part of the flat portion is the protrusion forming region **14***a*, a part of the center of the flat portion is preferably the protrusion forming region **14***a*.

[0055] In the present embodiment, as shown in FIGS. **6**A and **6**B, the protrusion **14***al* is provided at the uppermost end of the peripheral wall **14**, but is not limited thereto. The protrusion **14***a***1** may also be provided at an intermediate portion of the peripheral wall **14** in the depth direction of the attachment hole **12**, or may be provided at the lowermost end. In a case where the protrusion **14***a***1** is provided at the end portion (upper end portion or lower end portion) of the peripheral wall **14** in the depth direction of the attachment hole **12**, when the shaft member **210** is inserted into the attachment hole **12**, one end (for example, upper end) offset from the center in the depth direction of the peripheral wall **14** provided with the protrusion **14***al* in the depth direction of the attachment hole **12** is pushed outward of the attachment hole **12** by the shaft member **210**. As a result, the leaf spring portion **11** is expanded outward of the attachment hole **12** while being twisted so as to rotate with respect to the extending direction of the leaf spring portion **11** about the other end (for example, the lower end) of the leaf spring portion **11** to be described later in the depth direction of the attachment hole **12**. In this case, it is easy to bend the leaf spring portion **11** as compared with the case where the protrusion **14***a***1** is provided in the intermediate portion of the peripheral wall **14** in the depth direction of the attachment hole **12**. Further, in the peripheral wall **14**, the protrusion **14***a***1** is provided at the end portion (lower end portion in the present embodiment) on the side where the shaft member **210** enters when the shaft member **210** is inserted into the attachment hole **12**, so that the protrusion **14***a***1** can be brought into pressure contact with the side surface of the shaft member **210** even when the shaft member **210** is short. In addition, in the peripheral wall **14**, the protrusion **14***a***1** is provided at the end portion (the upper end portion in the present embodiment) opposite to the side where the shaft member **210** enters when the shaft member **210** is inserted into the attachment hole **12**, and the lower end portion is recessed more than the protrusion **14***a***1**, so that a sufficient dimension for inserting the shaft member **210** is ensured in the attachment hole **12**. Therefore, the shaft member **210** can be easily inserted into the attachment hole **12**. In addition, by providing the protrusion **14***a***1** at the upper end portion far from the installation surface **200**, a large moment can be applied to the shaft member **210**, and the protrusion **14*a*1** favorably comes into pressure contact with the side surface of the shaft member **210**.

[0056] In addition, a plurality of protrusions **14*al*** may be provided in the peripheral wall **14**. For example, as in a second attachment hole **12*b*** to be described later, two protrusions **14*a*1** may be disposed apart from each other in the circumferential direction of the peripheral wall **14**. Further, the plurality of protrusions **14*al*** may be arranged to be separated from each other in the z-axis direction. The circumferential direction of the peripheral wall **14** is an extending direction of the peripheral wall **14**, and is a direction when the peripheral wall **14** goes around the periphery of the attachment hole **12**.

[0057] Further, the entire protrusion forming region **14*a*** may be the protrusion **14*a*1**. For example, the protrusion forming region **14*a*** may be formed as an inclined surface that protrudes toward the inside of the attachment hole **12** from one end (for example, a lower end) to the other end (for example, an upper end) in the z-axis direction of the protrusion forming region **14*a***.

[0058] In the present embodiment, the protrusion **14*a*1** is long in the circumferential direction of the peripheral wall **14** (a direction that goes around the peripheral wall **14** orthogonal to the depth direction of the attachment hole **12**), and extends along the direction. That is, the protrusion **14*a*1** extends in a direction intersecting the depth direction of the attachment hole **12**. The protrusion **14*a*1** need not extend in the circumferential direction of the peripheral wall **14**. For example, the protrusion **14*al*** may extend in a spiral direction inclined to the same extent as a spiral groove **211** of the shaft member **210** to be described later with respect to the circumferential direction.

[0059] Since the protrusion **14*al*** extends in the direction intersecting the depth direction of the attachment hole **12**, when the shaft member **210** is inserted into the attachment hole **12**, the protrusion **14*a*1** comes into pressure contact with the side surface of the shaft member **210** by a surface or a line extending in the direction intersecting the axial direction of the shaft member **210**. Thus, even if the shaft member **210** tries to move in the depth direction of the attachment hole **12** (axial direction of the shaft member **210**), the shaft member **210** can be sufficiently held by the protrusion **14*a*1**.

[0060] The electronic component holder **100** in the present embodiment includes a case portion **20**. The case portion **20** is a member that encloses at least a part of the electronic component. In the present embodiment, the case portion **20** has a bottomed recess (electronic component housing recess **21**) that has an opening **21*a*** on the surface of the case portion **20** and into which an electronic component is inserted. Specifically, the case portion **20** has an electronic component housing recess **21** having an opening **21*a*** on a side surface facing the side (particularly, one side in the x-axis direction). Hereinafter, a direction from the bottom portion of the electronic component housing recess **21** toward the opening **21*a*** may be referred to as an opening direction. The electronic component is inserted into the electronic component housing recess **21** through the opening **21*a*** and housed in the case portion **20**. After the electronic component is stored in the electronic component housing recess **21**, the opening **21*a*** may be closed by a lid portion (not shown).

[0061] Instead of the present embodiment, the opening **21*a*** in the electronic component housing recess **21** may be opened in either the y-axis direction or the z-axis direction.

[0062] In the present embodiment, the case portion **20** is integrally formed of the same material as the flange portion **10**, but is not limited thereto. The case portion **20** and the flange portion **10** may be formed of another member and connected by adhesion, engagement, or the like.

[0063] In addition, instead of the present embodiment, the electronic component holder **100** may not include the case portion **20** that encloses the electronic component. For example, the electronic component may be attached to the base portion **10** in an exposed state.

[0064] The electronic component holder **100** of the present embodiment further includes a connector portion **30**. An external plug is attached to the connector portion **30** by insertion or the like, and an external device such as a control device and the electronic component are connected via the plug. In the present embodiment, the connector portion **30** has a bottomed recess (not shown in the drawings) that opens in a direction (x direction) opposite to the opening direction (−x direction) of the electronic component housing recess **21**. The external plug is inserted into the recess. The recess communicates with the electronic component housing recess **21** through a hole also received in the bottom, and the plug can be connected to the electronic component housed in the electronic component housing recess **21** through the hole.

[0065] In the present embodiment, the connector portion **30** is integrally formed of the same material as the case portion **20**, but the case portion **20** and the connector portion **30** may be formed as separate members and connected by adhesion, engagement, or the like.

[0066] The shaft member **210** is a member that is long in the axial direction and is inserted into the attachment hole **12**. In the present embodiment, the shaft member **210** is a metal bolt, but is not limited thereto. The shaft member **210** may be a rod-shaped member formed of a material other than metal, and may be made of resin, for example.

[0067] The fact that the cross section of the shaft member **210** is circular means that a cross section (end surface) obtained by cutting the shaft member **210** in an axis orthogonal direction is substantially circular. When the shaft member **210** is described as "cross section" without any description, it means a cross section. A cross section of the shaft member **210** is not limited to a perfect circular shape. For example, when a spiral groove is formed on the side surface of the shaft member **210** as described later, the cross section of the shaft member **210** may not be a perfect circle. Further, the outer edge of the cross section of the shaft member **210** does not need to smoothly continue, and may have a corner, for example, at the end point of the spiral. It is sufficient that the cross section has a circular shape to such an extent that the widths of the shaft member **210** in a plurality of arbitrary directions in the lateral direction (axis orthogonal direction) are equal to each other.

[0068] As shown in FIG. **6**B, a spiral groove (hereinafter, referred to as a spiral groove **211**) is formed on the entire side surface of the shaft member **210** in the present embodiment. Although an extending direction (hereinafter, also referred to as a spiral direction) of the spiral groove **211** is slightly inclined with respect to the circumferential direction of the shaft member, the spiral direction of the spiral groove **211** and the circumferential direction of the shaft member

210 approximately coincide with each other. An engagement groove 212 described later is a partial length region of the spiral groove 211.

[0069] The side surface of the shaft member 210 is a peripheral surface of the shaft member 210, and is a surface extending in the axial direction and the circumferential direction of the shaft member 210. A wall portion and a bottom portion constituting the spiral groove 211 and the engagement groove 212 constitute a part of the side surface of the shaft member 210.

[0070] In the present embodiment, the shaft member 210 is welded to the installation surface 200 as described later, but the shaft member 210 may be formed integrally with the installation surface 200 in advance. That is, the shaft member 210 may be a part of the installation surface 200. In the shaft member 210, one end connected to the installation surface 200 may be referred to as a proximal end, and one end opposite to the proximal end may be referred to as a distal end.

[0071] As described above, a distance between the protrusion 14*a*1 and the center of the attachment hole 12 is smaller than the radius of the shaft member 210 as viewed in the z-axis direction. The distance between the protrusion 14*al* and the center of the attachment hole 12 as viewed from the z-axis direction is the shortest distance between the protrusion 14*al* and the center of the attachment hole 12 in a natural state before the shaft member 210 is inserted (a state in which the attachment hole 12 is not expanded by the shaft member 210 as described later). The center of the attachment hole 12 having a rectangular shape like the first attachment hole 12*a* in the present embodiment is a center line bisecting the short direction (a line connecting points equidistant from two sides facing each other in the short direction among four sides constituting the first attachment hole 12*a*). That is, a distance between the center line of the first attachment hole 12*a* and the top portion of the protrusion 14*a*1 is smaller than the radius of the shaft member 210. When the attachment hole 12 has a regular polygon shape, the center point of the regular polygon is the center of the attachment hole 12, and when the shape of the attachment hole 12 is a triangle like the second attachment hole 12*b*, the inner center of the triangle is the center of the attachment hole 12 as described later.

[0072] As described later, the shaft member 210 is inserted into the attachment hole 12 in a direction in which the axial direction of the shaft member 210 coincides with the depth direction of the attachment hole 12. The shaft member 210 is inserted into the attachment hole 12 so deeply that the side surface of the shaft member 210 and the protrusion 14*al* are in contact with each other. A state in which the shaft member 210 is inserted into the attachment hole 12 and the protrusion 14*a*1 is in pressure contact with the side surface of the shaft member 210 is referred to as an attached state.

[0073] As described above, since the distance between the protrusion 14*a*1 and the center of the attachment hole 12 is smaller than the radius of the shaft member 210, the peripheral wall of the attachment hole 12 through which the shaft member 210 is inserted into the attachment hole 12 is pushed to the outside of the attachment hole 12 by the side surface of the shaft member 210, and the attachment hole 12 is expanded by the shaft member 210. As a result, in the attached state, the protrusion 14*a*1 (particularly, the top portion) is in pressure contact with the side surface of the shaft member 210 in a direction toward the axial center of the shaft member 210. More specifically, in the attached state, the top portion of the protrusion 14*al* and one or a plurality of facing portions described later are in pressure contact with the side surface of the shaft member 210. The shaft member 210 is biased by the peripheral wall 14 (the protrusion 14*a*1 and the facing portion 14*b*) at two or more positions facing each other with the attachment hole 12 interposed therebetween, so that the peripheral wall 14 holds the shaft member 210 at a predetermined position. That is, the relative positional relationship between the shaft member 210 and the attachment hole 12, and the relative positional relationship between the shaft member 210 and the electronic component holder 100 are fixed.

[0074] In the present embodiment, a part of the base portion 10 is a leaf spring portion 11 extending along the periphery of the attachment hole 12. A pressed surface (attachment hole facing surface 14*c*) of the leaf spring portion 11 is a part of the peripheral wall 14. Here, the pressed surface of the leaf spring portion 11 is a surface of the leaf spring portion 11 that is pressed and expanded by the shaft member 210 inserted into the attachment hole 12. The leaf spring portion 11 of the present embodiment has a thin plate shape, and one of a pair of main surfaces facing the inside of the attachment hole 12 is a pressed surface. Both ends of the leaf spring portion 11 in the extending direction of the leaf spring portion 11 are formed continuously with an adjacent portion 13 of the base portion 10 adjacent to the leaf spring portion 11. A central portion of the leaf spring portion 11 in the extending direction of the leaf spring portion 11 is separated from another part of the base portion 10 (the entire base portion 10 excluding the leaf spring portion 11, also referred to as a leaf spring outer portion). The leaf spring portion 11 can be bent so that the intermediate portion of the leaf spring portion 11 is displaced toward the center of the attachment hole 12 or opposite to the center of the attachment hole 12.

[0075] The leaf spring portion 11 can be bent in a direction intersecting (orthogonal to) the extending direction of the leaf spring portion 11 by applying an external force, and has a restoring force to return to a natural state (a state in which no external force is applied and there is no bending) in a bent state. Thus, in the attached state in which the shaft member 210 is inserted into the attachment hole 12, the leaf spring portion 11 is more favorably brought into pressure contact with the side surface of the shaft member 210 by the elastic restoring force of the leaf spring portion 11, and the shaft member 210 is firmly gripped by the leaf spring portion 11 and the facing portion 14*b* to be described later.

[0076] Here, the leaf spring portions 11 has a higher ease of elastic deformation in a direction away from the center of the attachment hole 12 (also referred to as outward) and a direction toward the center of the attachment hole 12 (also referred to as inward and also referred to as an inward and outward direction together with the outward direction) than in other directions orthogonal to the inward and outward directions. The leaf spring portion 11 is a member having a spring property that elastically deforms flexibly in the inward and outward directions.

[0077] As shown in FIG. 5A, the leaf spring portion 11 is long along the periphery of the attachment hole 12, and both ends in the extending direction of the leaf spring portion 11 are continuous with the adjacent portion 13 which is a part of the base portion 10 and adjacent to a proximal end portion of the leaf spring portion 11. The leaf spring portion 11 is connected to the leaf spring outer portion of the base portion **10** only at both ends in the extending direction of the leaf spring portion **11**. In other words, as described above, the intermediate portion in the extending direction of the leaf spring portion **11** is separated from the leaf spring outer portion (for example, a facing portion **14***b* to be described later or a through hole facing portion **16** to be described later). That is, in the leaf spring portion **11**, both end portions connected to the adjacent portion **13** are fixed, and the intermediate portion can freely move. In other words, the leaf spring portion **11** is a member supported at both ends by the base portion **10** (adjacent portion **13**), and its intermediate portion is capable of bending in a predetermined direction (in this embodiment, direction of the thickness of the leaf spring portion **11**). When an external force in a direction orthogonal to the extending direction of the leaf spring portion **11** is applied to the intermediate portion, the leaf spring portion **11** bends in an arcuate manner so that the intermediate portion (particularly, a part to which the external force is applied) moves outward with both end portions of the leaf spring portion **11** as fulcrums. Hereinafter, a direction in which the leaf spring portion **11** is bent (a direction in which the intermediate portion of the leaf spring portion **11** is displaced) is referred to as a bending direction.

[0078] Here, the fact that the leaf spring portion **11** can be bent means that the intermediate portion of the leaf spring portion **11** is displaced from the natural state by a force of an extent of manually inserting the shaft member **210** into the attachment hole **12**, the leaf spring portion **11** becomes arcuate, and the protrusion **14***al* is brought into a pressure contact state with the shaft member **210**. That is, the leaf spring portion **11** has sufficient flexibility to insert the shaft member **210** into the attachment hole **12**, and has sufficient flexibility to bring the protrusion **14***a***1** into pressure contact with the shaft member **210**. In the peripheral wall of the through hole (**3***fb*) in JP 2016-100610 A, one end in the depth direction of the through hole (**3***fb*) is connected to a part of the flange portion (**3***f*) (specifically, a bottom portion of a carved-out recess received also in the flange portion (**3***f*)), and the peripheral wall of the through hole (**3***fb*) does not have a sufficient spring property and is rigid. That is, the peripheral wall of the through hole (**3***fb*) in JP 2016-100610 A cannot be bent like the leaf spring portion **11**.

[0079] The dimension of the leaf spring portion **11** in the extending direction of the leaf spring portion **11** is preferably equal to or larger than a diameter of the shaft member **210**. More preferably, the dimension of the leaf spring portion **11** in the extending direction of the leaf spring portion **11** is 1.5 times or larger than the diameter of the shaft member **210**. Since the leaf spring portion **11** is sufficiently long in the extending direction of the leaf spring portion **11**, the leaf spring portion **11** has sufficient flexibility.

[0080] The dimension of the leaf spring portion **11** in the extending direction of the leaf spring portion **11** is preferably smaller than three times the diameter of the shaft member. In other words, in the attached state, a distance between the side surface of the shaft member **210** and the peripheral wall **14** is preferably smaller than the diameter of the shaft member **210**. Specifically, as shown in FIG. **5**A, in the first attachment hole **12***a* having a horizontally long rectangular shape, a distance in the x-axis direction between the peripheral wall facing in the x-axis direction and the side surface of the shaft member **210** is smaller than the diameter of the shaft member **210**. In the triangular second attachment hole **12***b*, a distance between the side surface of the shaft member **210** and a part of the peripheral wall **14** corresponding to a vertex of the triangle is smaller than the diameter of the shaft member **210**. As described above, since the shaft member **210** and the peripheral wall **14** are not largely separated from each other and the leaf spring portion **11** is not excessively long, the central portion (portion in pressure contact with the shaft member) of the leaf spring portion **11** can be in pressure contact with the shaft member **210** with a sufficient force for gripping the shaft member **210**.

[0081] In the present embodiment, the leaf spring portion **11** is linear. Since the leaf spring portion **11** is linear, the leaf spring portion **11** can be easily bent in the middle thereof. Here, the linear shape is not limited to a perfect linear shape, and may be slightly curved. Instead of the present embodiment, the leaf spring portion **11** may have an arc shape or a wave shape having an uneven shape as a whole.

[0082] As shown in FIG. **6**B, in the present embodiment, a lower surface of the leaf spring portion **11** is disposed on the same plane as the lower surface of the flange portion **10**, but is not limited thereto. The lower surface of the leaf spring portion **11** may be disposed on the inner side (upper side) of the flange portion **10** with respect to the lower surface of the flange portion **10**. In this case, the lower surface of the leaf spring portion **11** is not in contact with the installation surface **200** in the attached state, and the bending of the leaf spring portion **11** is not hindered by the friction between the leaf spring portion **11** and the installation surface **200**.

[0083] The leaf spring portion **11** is preferably a plate-like member in which a part of the pressed surface facing the attachment hole **12** (attachment hole facing surface **14***c* to be described later) is a main surface, and the main surface is larger than the side surface. Specifically, it is preferable that a dimension (hereinafter, also referred to as a width dimension of the leaf spring portion **11**) of the leaf spring portion **11** in the direction (bending direction, the y-axis direction in the first attachment hole **12***a*) orthogonal to the pressed surface (attachment hole facing surface **14***c*) of the leaf spring portion **11** that is a part of the peripheral wall **14** is smaller than a dimension of the leaf spring portion **11** in the depth direction (z-axis direction) of the attachment hole **12**. In such a leaf spring portion **11**, the bending direction of the leaf spring portion **11** can be said to be a direction perpendicular to the plane of the leaf spring portion **11** (in particular, the main surface). When the cross section of the leaf spring portion **11** is rectangular as in the present embodiment, as shown in FIG. **6**B, the z-axis direction is the longitudinal direction of the rectangular cross section, and the lateral direction is the lateral direction of the rectangular cross section. Unlike the present embodiment, when the cross section of the leaf spring portion **11** has another flat shape (for example, an elliptical shape), the largest dimension in the z-axis direction in the cross section is larger than the largest dimension in the lateral direction.

[0084] Since the leaf spring portion **11** is thin in the bending direction, the leaf spring portion **11** can have sufficient flexibility for bending. In addition, since the dimension of the leaf spring portion **11** in the z-axis direction is large, the elasticity of the leaf spring portion **11** is large as compared with the case where the dimension of the leaf spring portion **11** in the z-axis direction is small, and the restoring force for returning from a bent state to a natural state is large. That is, in the attached state, the leaf spring portion **11** (in particular, the protrusion **14***a***1**) comes into pressure contact with the shaft member with a stronger force.

[0085] In the present embodiment, the dimension in the bending direction is smaller than the dimension in the z-axis direction over the entire length in the extending direction of the leaf spring portion **11**, but is not limited thereto. The dimension in the bending direction may be smaller than the dimension in the z-axis direction only in a partial length region in the extending direction of the leaf spring portion **11**. That is, only a partial length region of the long member may have sufficient flexibility and function as the leaf spring portion **11**. It is desirable that an intermediate portion of the long member in which at least the protrusion **14***a***1** is provided has a dimension in the bending direction smaller than a dimension in the z-axis direction and functions as the leaf spring portion **11**.

[0086] Instead of the present embodiment, when the flange portion **10** is thin, the dimension of the leaf spring portion **11** in a direction orthogonal to the pressed surface of the leaf spring portion **11** which is a part of the peripheral wall **14** may be larger than a dimension of the leaf spring portion **11** in the depth direction of the attachment hole **12**. For example, the area of the surface (end surface) of the leaf spring portion **11** oriented in the z-axis direction may be larger than the area of the pressed surface of the leaf spring portion **11** which is a part of the peripheral wall **14**. With this configuration, since the leaf spring portion **11** has a sufficiently large dimension in the bending direction, it is possible to prevent the leaf spring portion **11** from being accidentally broken when the leaf spring portion **11** is bent. Even in such a configuration, the leaf spring portion **11** has sufficient flexibility for inserting the shaft member **210** into the attachment hole **12**.

[0087] As shown in FIG. **5**A, the facing portion **14***b* which is a part of the peripheral wall **14** faces the pressed surface (attachment hole facing surface **14***c*) of the leaf spring portion **11** across the attachment hole **12**. At least a part of the attachment hole facing surface **14***c* of the leaf spring portion **11** is a forming region of the protrusion (protrusion forming region **14***a*). In other words, the facing portion **14***b* faces the protrusion forming region **14***a* including the protrusion **14***a***1**. The facing portion **14***b* is a non-forming region of the protrusion **14***a***1**.

[0088] As described later, when the shaft member **210** is inserted into the attachment hole **12**, the peripheral wall **14** is expanded while the shaft member **210** pushes the protrusion **14***a***1** outward. Since the protrusion **14***a***1** is provided in the leaf spring portion **11** having flexibility, when the protrusion **14***a***1** is pushed outward by the insertion of the shaft member **210**, the leaf spring portion **11** including the protrusion **14***a***1** is bent, and the attachment hole **12** is easily widened as much as the shaft member **210** can be inserted. As compared with a case where the protrusion **14***a***1** is provided in a portion other than the leaf spring portion **11** (a portion having lower flexibility than the leaf spring portion **11**), the shaft member **210** can be easily inserted into the attachment hole **12**. Further, when the protrusion **14***a***1** is pushed, the leaf spring portion **11** including the protrusion is bent, so that the protrusion **14***al* is prevented from being damaged by the force of the shaft member **210** pushing the protrusion **14***a***1**.

[0089] Since the facing portion **14***b* does not have the protrusion **14***a***1** and is a flat surface at least in the z-axis direction, the side surface of the shaft member **210** and the peripheral wall **14** (facing portion **14***b*) are brought into pressure contact with each other at a plurality of points continuous in the z-axis direction as described later, and the peripheral wall **14** can grip the shaft member **210** more firmly. Further, since the peripheral wall **14** supports the shaft member **210** at a plurality of places continuous in the z-axis direction, it is possible to prevent the shaft member **210** from being inclined with respect to the depth direction of the attachment hole **12** in the attached state.

[0090] The facing portion **14***b* is a part of the peripheral wall **14** that faces the leaf spring portion **11** (particularly, an intermediate portion thereof) with the attachment hole **12** interposed therebetween. As described later, when the shaft member **210** is inserted into the attachment hole **12**, the side surface of the shaft member **210** is in contact with the facing portion **14***b*. The facing portion **14***b* is preferably a gently curved surface or a flat surface. For example, as shown in FIG. **5**A, the first attachment hole **12***a* has a horizontally long rectangular shape, but the facing portion **14***b* is a part of the peripheral wall **14** that faces the attachment hole facing surface **14***c* by 180 degrees and extends in the longitudinal direction (x-axis direction). As shown in FIG. **7**A, the second attachment hole **12***b* has an equilateral triangle shape and has two leaf spring portions **11***a* and **11***b*. The facing portion **14***b* that faces the attachment hole facing surface **14***cl* in the leaf spring portion **11***a* of the second attachment hole **12***b* is a portion facing the attachment hole facing surface **14***c* at 120 degrees with the attachment hole **12** interposed therebetween, and specifically, is one flat portion constituting an equilateral triangle of the peripheral wall **14**. Note that, when the leaf spring portion **11***b* is not provided, a flat portion corresponding to an attachment hole facing surface **14***c***2** in the leaf spring portion **11***b* in FIG. **7**A may also be the facing portion **14***b*.

[0091] As described above, the entire attachment hole facing surface **14***c* may be the protrusion forming region **14***a*, and only a part of the attachment hole facing surface **14***c* may be the protrusion forming region **14***a*. The protrusion **14***al* may be formed at least in a middle region of the attachment hole facing surface **14***c* in the extending direction of the leaf spring portion **11** (particularly, a portion with which the shaft member **210** comes into contact or approaches in the attached state).

[0092] The facing portion **14***b* being a non-forming region of the protrusion **14***a***1** means that the non-forming region is a surface having substantially no unevenness. That is, as described above, the facing portion **14***b* is a substantially flat surface or a gently curved surface. The gently curved surface means that it is gentler than the arc shape of the outer edge of the shaft member **210**.

[0093] As shown in FIG. **6**B, in the attached state, the side surface of the shaft member **210** and the facing portion **14***b* are in contact with each other at a plurality of positions continuous in the z-axis direction. Specifically, the end portion (the outermost end portion in the radial direction of the shaft member **210**) of the wall portion forming the spiral groove **211** formed in the side surface of the shaft member **210** and the facing portion **14***b* are in contact with each other. Instead of the present embodiment, the side surface of the shaft member **210** and the facing portion **14***b* may be in substantial line or surface contact with each other. At this time, a contact surface (or a substantial line) between the side surface of the shaft member **210** and the facing portion

14*b* is along the z-axis direction. For example, when the spiral groove **211** is not provided in the shaft member **210**, or when the shaft member **210** and the facing portion **14***b* are in pressure contact with each other so that the side surface of the shaft member **210** bites into the spiral groove **211**, the side surface of the shaft member **210** and the facing portion **14***b* are substantially in line or plane contact with each other.

[0094] As shown in FIG. **5**A, the base portion **10** includes a facing wall portion **15** which includes the facing portion **14***b* and is positioned on a side away from the center of the attachment hole **12**. The dimension (width dimension) of the facing wall portion **15** in the direction orthogonal to the facing portion **14***b* is larger than the dimension (width dimension of the leaf spring portion **11**) of the leaf spring portion **11** in the direction orthogonal to the pressed surface (attachment hole facing surface **14***c*). More preferably, the width dimension of the facing wall portion **15** is larger than the dimension of the facing wall portion **15** in the z-axis direction. That is, the facing wall portion **15** is a non-forming region of the leaf spring portion **11**, and the facing wall portion **15** is a rigid member having substantially no flexibility to bend outward from the center of the attachment hole **12**.

[0095] The facing wall portion **15** is a part of the base portion **10** including the facing portion **14***b* and up to the outer edge of the base portion **10**. That is, the width dimension of the facing wall portion **15** refers to a distance from the facing portion **14***b* to the outer edge of the base portion **10**. For example, as shown in FIG. **5**A, in the first attachment hole **12***a*, a shortest distance from the facing portion **14***b* to the edge of the flange portion **10** located on a lower side of the facing portion **14***b* in the plane of drawing is the width dimension of the facing wall portion **15**. Further, as shown in FIG. **7**A, in the second attachment hole **12***b*, a shortest distance from the facing portion **14***b* to the proximal end of the flange portion **10** (the boundary between the flange portion **10** and the case portion **20**) is the width dimension of the facing wall portion **15**. Unlike the present embodiment, when the second attachment hole **12***b* is disposed on the more proximal end side of the flange portion **10** (the right side in FIG. **7**A) than the present embodiment, the dimension of a part of the case portion **20** (a part of the case portion **20** on the flange portion **10** side) may also be included in the width dimension of the facing wall portion **15**. This is because, as described above, a part of the case portion **20** on the flange portion **10** side is located near the second attachment hole **12***b*, and this part is also the base portion **10**.

[0096] Since the facing wall portion **15** is rigid, the shaft member **210** can be firmly gripped by the leaf spring portion **11** and the facing wall portion **15**. Further, since the facing wall portion **15** is rigid and does not deform, the electronic component holder **100** is disposed so that the facing wall portion **15** is in contact with the side surface of the shaft member **210** in the attached state. Thus, the shaft member **210** and the electronic component holder **100** can be fixed in a desired positional relationship at least in a direction orthogonal to the facing portion **14***b*.

[0097] As shown in FIG. **5**A, the base portion **10** has a through hole **18** provided in the vicinity of the attachment hole **12** along a part of the peripheral wall **14**. The part of the peripheral wall **14** along which the through hole **18** extends is a pressed surface (attachment hole facing surface **14***c*) of the leaf spring portion **11** facing the attachment hole **12**. In the present embodiment, the through hole **18** is formed along one of the flat portions of the peripheral wall **14** defining the polygonal attachment hole **12**. For example, the through hole **18** provided in the first flange portion **10***a* is formed along a flat portion of the peripheral wall **14** forming a long side of the first attachment hole **12***a* having a horizontally long rectangular shape.

[0098] The leaf spring portion **11** is disposed between the attachment hole **12** and the through hole **18**. That is, a part of the base portion **10** formed between the attachment hole **12** and the through hole **18** serves as the leaf spring portion **11**. Arranging the through hole **18** and the attachment hole **12** in the vicinity means that a part of the base portion **10** sandwiched between the through hole **18** and the attachment hole **12** is sufficiently close to have flexibility as the leaf spring portion **11**. Specifically, a distance between the attachment hole **12** and the through hole **18** (the width dimension of the leaf spring portion **11**) is preferably smaller than the dimension of the leaf spring portion **11** in the depth direction of the attachment hole **12**.

[0099] When the through hole **18** is provided, as compared with the case where a peripheral edge portion of the flange portion **10** is the leaf spring portion **11** without providing the through hole **18** as described in the modification described later, even if the leaf spring portion **11** is accidentally damaged, the electronic component holder **100** can be prevented from completely detaching from the attachment destination member. Specifically, for example, when the leaf spring portion **11** is lost due to breakage, it is difficult to hold the shaft member **210** by the leaf spring portion **11** and the facing portion, but the shaft member **210** is still surrounded by the inner wall of the flange portion **10** defining the attachment hole **12** and the through hole **18**. Therefore, the shaft member **210** is prevented from completely coming off from the attachment hole **12**.

[0100] A dimension in a direction orthogonal to the extending direction of the through hole **18** in a part of the base portion **10** (the through hole facing portion **16** shown in FIG. **5**A) facing the leaf spring portion **11** with the through hole **18** interposed therebetween is preferably larger than the width dimension of the leaf spring portion **11**. Since rigidity of the part of the leaf spring portion **11** in the bending direction is larger than rigidity of the leaf spring portion **11** in the bending direction, even when the leaf spring portion **11** is accidentally lost as described above, it is possible to prevent the shaft member **210** from being detached from the flange portion **10** due to the loss of the part.

[0101] In the present embodiment, the through hole **18** is long in the extending direction of the leaf spring portion **11**. The extending direction of the leaf spring portion **11** and the extending direction of the through hole **18** are preferably the same direction. A width of the through hole **18** (dimension in a direction orthogonal to the depth direction of the attachment hole **12** and the extending direction of the through hole **18**) is smaller than a dimension of the leaf spring portion **11** (width dimension of the leaf spring portion **11**) in a direction orthogonal to the attachment hole facing surface **14***c*. Further, a width dimension of the through hole **18** is preferably larger than the amount by which the leaf spring portion **11** is displaced in the bending direction in the attached state. In the present embodiment, the width of the through hole **18** is larger than the protrusion height of the protrusion **14**a**1** (the height of the top surface **14**a**2** with respect to the recessed surface **14**a**4**).

[0102] By limiting the width of the through hole **18** to be small, it is possible to prevent the leaf spring portion **11** from being excessively bent and damaged beyond its own flexibility. This is because even if the leaf spring portion **11** attempts to bend excessively, the leaf spring portion **11** abuts on the through hole facing portion **16**, and the leaf spring portion **11** cannot bend more largely than the width of the through hole **18** in the bending direction.

[0103] Furthermore, by reducing the width of the through hole **18**, the flange portion **10** can be reduced.

[0104] Instead of the present embodiment, the width dimension of the through hole **18** may be larger than the width dimension of the leaf spring portion **11**. In this case, since the through hole **18** has a large dimension in the bending direction of the leaf spring portion **11**, the leaf spring portion **11** can be greatly bent. That is, even when the distance between the center of the attachment hole **12** and the protrusion **14**a**1** is made sufficiently smaller than the radius of the shaft member **210**, and the leaf spring portion **11** is largely bent to grip the shaft member **210** with a stronger force, the leaf spring portion **11** and the through hole facing portion **16** do not interfere with each other.

[0105] As shown in FIGS. **6**A and **6**B, a top portion of the protrusion **14**a**1** protruding most toward the inside of the attachment hole **12** is a flat top surface **14**a**2** extending in the depth direction of the attachment hole **12**. That is, the protrusion **14**a**1** has a thickness of a predetermined dimension (dimension of the top surface **14**a**2** in the depth direction of the attachment hole **12**) or more in the depth direction of the attachment hole **12**. In the attached state, the top surface **14**a**2** faces the side surface of the shaft member **210**. When the shaft member **210** is inserted into the attachment hole **12**, it is rubbed against the shaft member **210**, so that the protrusion **14**al may be worn, that is, the protrusion **14**a**1** may become thin in the depth direction of the attachment hole **12**. Since the protrusion **14**a**1** has a predetermined dimension in the depth direction of the attachment hole **12**, even if the protrusion **14**a**1** wears due to the shaft member **210** being inserted into the attachment hole **12**, the protrusion **14**a**1** can have a sufficient protruding height to be in pressure contact with the shaft member **210**.

[0106] Instead of the present embodiment, the top portion of the protrusion **14**al may be a corner portion. That is, the top portion may be a substantial line along the circumferential direction of the peripheral wall **14**. When the top portion is a corner portion, it is possible to make pressure contact so as to bite into the side surface of the shaft member **210** (particularly, a bottom portion **212**b of the engagement groove **212**).

[0107] As shown in FIG. **6**B, the side surface of the shaft member **210** is provided with a bottomed groove portion (engagement groove **212**) that engages with the protrusion **14**a**1** when the shaft member **210** is inserted into the attachment hole **12**. The dimension of the top surface **14**a**2** in the depth direction of the attachment hole **12** (the width dimension of the top surface **14**a**2**) is larger than the width of the bottom portion **212**b of the engagement groove **212** in the axial direction of the shaft member **210**. The engagement groove **212** is a groove provided on the side surface of the shaft member **210** to engage with the protrusion **14**a**1** in the attached state. In the present embodiment, a partial length region of the spiral groove **211** is the engagement groove **212**. Specifically, in the spiral groove **211**, a partial length region facing the protrusion **14**a**1** in the attached state is the engagement groove **212**. As will be described later, the engagement groove **212** may extend in a direction intersecting the axial direction of the shaft member **210** and have a constant length, or may be a short groove.

[0108] As shown in FIG. **6**C, the engagement groove **212** is defined by the bottom portion **212**b and a pair of wall portions facing each other in the axial direction of the shaft member **210**. In the present embodiment, the bottom portion **212**b of the engagement groove **212** has a planar shape having a certain dimension in the axial direction of the shaft member **210**, but the bottom portion **212**b may have a linear shape without having a substantial length in the axial direction of the shaft member **210**.

[0109] Since the width dimension of the top surface **14**a**2** is larger than the width of the bottom portion of the engagement groove **212**, the protrusion **14**a**1** comes into pressure contact with each of the pair of wall portions defining the engagement groove **212** in the attached state as described later. Thus, the shaft member **210** can be more firmly gripped by the protrusion **14**al and the facing portion **14**b. Further, since the protrusion **14**a**1** is in pressure contact with both of the pair of wall portions facing each other in the depth direction of the attachment hole **12**, the shaft member **210** and the electronic component holder **100** are prevented from being relatively displaced in the z-axis direction in the attached state.

[0110] Instead of the present embodiment, the width dimension of the top surface **14**a**2** may be equal to or smaller than the width of the bottom portion **212**b of the engagement groove **212**. In this case, in the attached state, the top surface **14**a**2** and the bottom portion **212**b of the engagement groove **212** may be in pressure contact with each other. In this case, since the contact surface between the protrusion **14**a**1** and the side surface of the shaft member **210** becomes large, the relative positional relationship between the shaft member **210** and the electronic component holder **100** is prevented from being deviated in the z-axis direction by friction of the contact surface.

[0111] The engagement groove **212** extends in a direction intersecting the axial direction of the shaft member **210**. In the present embodiment, the engagement groove **212** extends in the spiral direction of the spiral groove **211**. Alternatively, the engagement groove **212** may extend in a direction orthogonal to the axial direction of the shaft member **210**. At least one wall portion (pressure contact wall portion **212**c) of the two wall portions forming the engagement groove **212** so as to sandwich the engagement groove **212** in the axial direction of the shaft member **210** is inclined with respect to the axial direction of the shaft member **210**. Among the pair of wall portions, at least a wall portion (lower wall portion in the present embodiment) close to the attachment destination member (installation surface **200**) is preferably inclined with respect to the axial direction. In the present embodiment, both of the pair of wall portions are inclined with respect to the axial direction of the shaft member **210**, but the lower wall portion is referred to as a pressure contact wall portion **212**c.

[0112] The protrusion **14**al has an inclined surface **14**a**3** that is inclined with respect to the depth direction of the attachment hole **12** and faces one wall portion when the shaft member **210** is inserted into the attachment hole **12**. The inclined surface **14**a**3** is disposed between the top surface

14*a*2 and the recessed surface 14*a*4. An inclination of the inclined surface 14*a*3 in the protrusion 14*a*1 with respect to the depth direction of the attachment hole 12 is smaller than an inclination of one wall portion (pressure contact wall portion 212*c*) of the engagement groove 212 with respect to the axial direction of the shaft member 210. Here, the inclination of the inclined surface 14*a*3 with respect to the depth direction of the attachment hole 12 and the inclination of the pressure contact wall portion 212*c* with respect to the axial direction are angles that are acute angles among angles formed by each direction and each plane. Since the angle of the inclined surface 14*a*3 with respect to the depth direction of the attachment hole 12 is larger than the angle with respect to the axial direction of the pressure contact wall portion 212*c*, the pressure contact wall portion 212*c* and the inclined surface 14*a*3 interfere with each other in the attached state. That is, in the attached state, the pressure contact wall portion 212*c* and the inclined surface 14*a*3 are in surface contact with each other and are in pressure contact with each other. The entire pressure contact wall portion 212*c* may be in surface contact with the inclined surface 14*a*3, or only a part of the pressure contact wall portion 212*c* on the outer side may be in surface contact with the inclined surface 14*a*3. However, as described later, in FIGS. 6B and 6C, for convenience, the protrusion 14*a*1 is shown in a state of interfering with the side surface of the shaft member 210 without being deformed.

[0113] The shaft member 210 is prevented from being detached from the attachment hole 12 by friction between the pressure contact wall portion 212*c* and the inclined surface 14*a*3 due to pressure contact between the pressure contact wall portion 212*c* and the inclined surface 14*a*3 at the surfaces.

[0114] The protrusion 14*al* has an end surface 11*c*1 connected to the opposite side of the inclined surface 14*a*3 with respect to the top portion. The end surface 11*c*1 faces a direction different from the inclined surface 14*a*3 in the depth direction. The inclination of the end surface 11*cl* of the protrusion 14*a*1 with respect to the depth direction of the attachment hole 12 is larger than the inclination of the inclined surface 14*a*3 of the protrusion 14*al* with respect to the depth direction of the attachment hole 12. That is, surfaces (the end surface 11*cl* and the inclined surface 14*a*3) of the protrusion 14*a*1 respectively facing up and down directions are arranged at respective different angles with respect to the z-axis direction. Specifically, the inclined surface 14*a*3 is a gentle slope with respect to the recessed surface 14*a*4, and the end surface 11*c*1 stands upright with respect to the axial direction of the shaft member 210 in the attached state. Since the inclined surface 14*a*3 and the recessed surface 14*a*4 smoothly continue, it is easy to insert the shaft member 210 into the attachment hole 12. In addition, since the angle of the end surface 11*c*1 with respect to the axial direction of the shaft member 210 in the attached state is large, the protrusion 14*al* is prevented from being displaced from the engagement groove 212 to an upper side of the engagement groove 212, and the shaft member 210 is prevented from coming out of the attachment hole 12. The end face 11*cl* is a surface facing the upper side among the surfaces of the protrusion 14*al*. In the present embodiment, the end face 11*c*1 extends in a direction orthogonal to the z-axis direction, but may be inclined with respect to the z-axis direction. The inclination of the end surface 11*c*1 with respect to the depth direction of the attachment hole 12 refers to an angle that is an acute angle among angles formed by the direction and the plane.

[0115] As shown in FIG. 3, in the present embodiment, the attachment hole 12 has a polygonal shape. In the present embodiment, the first attachment hole 12*a* has a horizontally long rectangular shape. A length of one side (long side) of the attachment hole 12 is larger than a length of the other side, and the leaf spring portion 11 is formed including the one side (long side). By providing the leaf spring portion 11 along the long side, the length of the leaf spring portion 11 can be ensured sufficiently large, and the flexibility of the leaf spring portion 11 can be increased. The polygon is not limited to a triangle and a quadrangle (including a rectangle and a square), and may be a polygon having five or more corners. The shape of the attachment hole 12 is preferably a triangle or a quadrangle. This is because one side of a triangle or a quadrangle is longer than that of a polygon having five or more corners, and a long leaf spring portion 11 can be provided along the one side. Since the leaf spring portion 11 is long, the leaf spring portion 11 has large flexibility.

[0116] As shown in FIG. 4, the opening 21*a* is disposed on one side (upper side) of the base portion (the first flange portion 10*a* in the present embodiment) in the depth direction of the attachment hole 12. In addition, the attachment hole 12 is disposed outside the case portion 20 (on the right side in FIG. 4) in the direction from the bottom portion of the electronic component housing recess 21 toward the opening 21*a*. As shown in FIG. 6C, an outer surface (end surface 11*c*1) of the protrusion 14*a*1 facing one side (upper side) is flush with a main surface of the base portion facing one side (upper side). The outer surface of the protrusion 14*a*1 facing one side refers to a part of the surface of the protrusion 14*a*1 that can be visually recognized from above. Here, the two surfaces being flush means that the two surfaces are located on the same plane at substantially the same height. That is, the end surface 11*c*1 and an upper surface of the flange portion 10 (including a leaf spring portion upper surface 11*c* of the leaf spring portion 11) are located on the same plane at the same height in the z-axis direction. In other words, the attachment hole 12 is formed to have an equal width or a wide width downward from the upper surface of the flange portion 10, and is not formed to have a width that is narrower downward from the upper surface of the flange portion 10.

[0117] Since the upper surface of the flange portion 10 disposed on the opening side of the electronic component housing recess 21 is flush with the end surface 11*c*1, the electronic component holder 100 can be easily manufactured by resin injection molding. This is because the upper surface of the flange portion 10 is flush with and flat with the end surface 11*c*1, whereby the upper surface can be a sliding surface of a slide mold, and thus the electronic component housing recess 21 and the flange portion 10 can be molded by ae slide mold having a simple structure. Instead of the present embodiment, when the end surface 11*c*1 is not flush with the upper surface of the flange portion 10 and is formed to be recessed from the upper surface, the slide mold may be a multi-stage drive type, and the slide mold may be removed from the electronic component housing recess 21 after opening the mold that has formed the recessed end surface 11*c*1.

[0118] Hereinafter, the second flange portion 10*b* and the second attachment hole 12*b* will be described in detail with reference to FIGS. **7**A to **8**B. The second flange portion **10***b* has a leaf spring portion **11** and a through hole **18** having features similar to those of the first flange portion **10***a* described above. The second flange portion **10***b* is different from the first flange portion **10***a* in the shape of the attachment hole **12** as viewed from the z-axis direction and in that a plurality of (two) leaf spring portions **11** and through holes **18** is provided.

[0119] As described above, the second attachment hole **12***b* has an equilateral triangular shape when viewed from the z-axis direction. In the attached state, the shaft member **210** is in pressure contact with each plane portion constituting each side of an equilateral triangle of the peripheral wall **14** defining the second attachment hole **12***b*.

[0120] As shown in FIG. **8**A, in the attached state, a part of the peripheral wall **14** facing the protrusion **14***a***1** provided in the leaf spring portion **11***a* at 180 degrees with the second attachment hole **12***b* interposed therebetween is not in contact with the side surface of the shaft member **210**. As shown in FIG. **8**A, the protrusion **14***a***1** is engaged with the engagement groove **212**. The aspect of engagement between the protrusion **14***a***1** and the engagement groove **212** is similar to the aspect of engagement between the protrusion **14***a***1** and the engagement groove **212** in the first attachment hole **12***a* shown in FIG. **6**C. As shown in FIG. **8**B, a part of the peripheral wall **14** facing the facing portion **14***b* at 180 degrees with the second attachment hole **12***b* interposed therebetween is not in contact with the side surface of the shaft member **210**.

[0121] In the second attachment hole **12***b*, the leaf spring portions **11** (leaf spring portions **11***a* and **11***b*) are provided along a part of the peripheral wall **14** forming at least two adjacent sides of the attachment hole **12**. Each of the leaf spring portions **11***a* and **11***b* is provided with a protrusion **14***a***1**. As viewed in the z-axis direction, a distance from each of the protrusions **14***a***1** to the center of the second attachment hole **12***b* (the inner center of a triangle in the present embodiment) is smaller than the radius of the shaft member **210**. Therefore, both the leaf spring portions **11***a* and **11***b* are in pressure contact with the shaft member **210**. Since the side surface of the shaft member **210** is biased in two directions (a direction orthogonal to an extending direction of the leaf spring portion **11***a* and a direction orthogonal to an extending direction of leaf spring portion **11***b*) by the two leaf spring portions **11**, the shaft member **210** can be prevented from moving with respect to the second attachment hole **12***b* in the two directions.

[0122] In addition, the second flange portion **10***b* has a facing portion **14***b* opposed to the pressed surface of the leaf spring portion **11***a* or the leaf spring portion **11***b*. At least a part (all in the present embodiment) of the pressed surfaces of the leaf spring portion **11***a* and the leaf spring portion **11***b* is the protrusion forming region **14***a*, and the facing portion **14***b* is the protrusion non-forming region. Furthermore, the width dimension of the facing wall portion **15** which includes the facing portion **14***b* in the second flange portion **10***b* and is located on a side away from the center of the attachment hole **12** is larger than a width dimension of the leaf spring portion **11***a* or the leaf spring portion **11***b*.

[0123] Here, as described above, the facing portion **14***b* in the second attachment hole **12***b* is a surface of the peripheral wall **14** facing the pressed surfaces of the leaf spring portions **11***a* and **11***b* at 120 degrees with the second attachment hole **12***b* interposed therebetween.

[0124] Further, two through holes **18** are provided in the vicinity of the second attachment hole **12***b*. Each of the two leaf spring portions **11***a* and **11***b* is disposed between the second attachment hole **12***b* and the through hole **18**.

[0125] Instead of the present embodiment, the shape of the second attachment hole **12***b* as viewed from the z-axis direction may be a square, and the leaf spring portion **11** may be provided on the peripheral wall **14** constituting two adjacent sides of the square. In this case, the electronic component holder **100** can be positioned with respect to the shaft member **210** in the bending direction of one leaf spring portion **11** and the bending direction of the other leaf spring portion **11**, that is, in two orthogonal directions.

(Electronic Component Holder)

[0126] The electronic component holder **100** according to the present embodiment can be provided only with the electronic component holder **100** without including the shaft member **210**.

[0127] As described above, the electronic component holder **100** is a vehicle electronic component holder used together with the shaft member **210** having a circular cross section. The electronic component holder **100** has a base portion **10** provided with the attachment hole **12** for inserting the shaft member **210**. The base portion **10** has the protrusion **14***a***1** protruding toward the inside of the attachment hole **12** from the peripheral wall **14** defining the attachment hole **12**. By using the electronic component holder **100** together with the shaft member **210** described above, it is possible to attach the electronic component to the attachment destination member without labor of holding the electronic component holder at the time of screw fastening.

(Method for Mounting Electronic Component)

[0128] Hereinafter, a method for attaching an electronic component to another member using the electronic component holder set of the present embodiment (hereinafter, the method may be referred to as the present method) will be described. Hereinafter, the method for attaching the electronic component may be described using a plurality of steps described in order, but the order described does not limit the order or timing of executing the plurality of steps.

[0129] Therefore, when the attachment method is performed, the order of the plurality of steps can be changed within a range in which there is no problem in terms of content, and a part or all of the execution timings of the plurality of steps may overlap each other.

[0130] As shown in FIG. **4**, the shaft member **210** is attached by welding, bonding, or the like so as to protrude from the attachment destination member. In FIGS. **4**, **6**B, **8**A, and **8**B, a mode of joining the shaft member **210** and the installation surface **200** is not shown, and the shaft member **210** and the installation surface **200** are shown as the same member. When the plurality of attachment holes **12** is provided in the electronic component holder **100**, the same number of shaft members **210** as the number of attachment holes **12** provided in the electronic component holder **100** are attached at positions spaced apart corresponding to the distance between the attachment holes **12**. In the electronic component holder **100** of the present embodiment, since the two attachment holes **12** are provided, the two shaft members **210** are attached to positions separated from the installation surface **200**.

[0131] An electronic component is inserted into the case portion **20** of the electronic component holder **100**. The electronic component may be inserted into the case portion **20** before or after the electronic component holder **100** is attached to the installation surface **200**.

[0132] The shaft member **210** is inserted into the attachment hole **12** in a state where the electronic component holder **100** maintains the inclination in which the axial direction of the shaft member **210** coincides with the depth direction of the attachment hole **12**. As shown in FIG. **6**B, in the present embodiment, the outer diameter of one end (tip) of the shaft member **210** is smaller than the lower width of the attachment hole **12**. More specifically, the outer diameter of the tip of the shaft member **210** is smaller than the distance between the recessed surface **14*a*4** of the attachment hole **12** and a part of the peripheral wall **14** facing the recessed surface **14*a*4** at 180 degrees. In addition, the distal end portion of the shaft member **210** gradually increases in diameter from the most distal end toward the proximal end side. Thus, the shaft member **210** can be easily inserted into the attachment hole **12**. In a case where the spiral groove **211** is formed on the side surface of the shaft member **210**, the protrusion **14*a*1** is engaged with the spiral groove **211** in the process of the shaft member **210** being inserted into the attachment hole **12**, and is repeatedly detached from the spiral groove **211**. The shaft member **210** is inserted into the attachment hole **12** until the protrusion **14*a*1** is engaged with the engagement groove **212**. In the attached state, the installation surface **200** and the lower surface of the flange portion **10** may be in contact with each other or may be partially separated from each other.

[0133] The method of attaching the electronic component to the installation surface **200** by the electronic component holder **100** and the shaft member **210** is not limited to the above-described method. For example, with the shaft member **210** attached to the electronic component holder **100** in advance, the shaft member **210** may be attached to the installation surface **200** by adhesion, engagement, or the like. Alternatively, with the electronic component holder **100** disposed on the installation surface **200**, the shaft member **210** may be inserted into the attachment hole **12** and a screw hole provided in the installation surface **200**.

[0134] In the attached state, the protrusion **14*al*** and the engagement groove **212** are engaged. In FIG. **6**C, the protrusion **14*a*1** and the side surface of the shaft member **210** are shown as interfering with each other, but actually, a part of the protrusion **14*a*1** or a part of the side surface of the shaft member **210** is scraped or deformed, and the pressure contact wall portion **212*c*** and the inclined surface **14*a*3** are engaged in a pressure contact state.

[0135] In FIGS. **6**B, **6**C, and **8**A, displacement of the leaf spring portion **11** in the bending direction is not shown in the drawings. In practice, the leaf spring portion **11** is displaced outward of the attachment hole **12** (rightward in the plane of drawing in FIG. **6**B) when the protrusion **14*a*1** is biased by the side surface of the shaft member **210**. In the present embodiment, since the protrusion **14*a*1** is provided at the upper end of the peripheral wall **14** as shown in FIG. **6**B, in particular, an upper end portion of the leaf spring portion **11** is displaced rightward in the plane of drawing. A lower end portion of the leaf spring portion **11** may or may not be displaced rightward in the plane of drawing together with the upper end portion. Alternatively, when an upper end portion of the leaf spring portion **11** is displaced rightward, the leaf spring portion **11** may rotate with respect to the extending direction of the leaf spring portion **11**, and the lower end portion of the leaf spring portion **11** may be displaced leftward in the drawing. In other words, in the attached state, the leaf spring portion **11** is inclined with respect to the axial direction of the shaft member **210** (the depth direction of the attachment hole **12**). Specifically, the leaf spring portion **11** is inclined away from the shaft member **210** from the lower end toward the upper end of the shaft member **210**. The lower end portion of the leaf spring portion **11** and the side surface of the shaft member **210** may be simply in contact with each other, may be in pressure contact with each other, or may be separated from each other. When the lower end portion of the leaf spring portion **11** is in contact with or pressed against the side surface of the shaft member **210**, the base portion **10** is securely fixed to the shaft member **210** at both the upper and lower ends of the leaf spring portion **11**. On the other hand, when the lower end portion of the leaf spring portion **11** is spaced apart from the side surface of the shaft member **210**, sufficient room remains for the upper end portion of the leaf spring portion **11** (protrusion **14*a*1**) to exert a biasing force against the shaft member **210**. Therefore, the base portion **10** is securely fixed to the shaft member **210** in either case.

[0136] As described above, the extending direction of the protrusion **14*a*1** is the direction orthogonal to the z-axis direction (the circumferential direction of the peripheral wall **14**), while the extending direction of the spiral groove **211** is not orthogonal to the z-axis direction. In the attached state, it is preferable that the protrusion **14*a*1** and the engagement groove **212** are in line or surface contact with each other by deformation or displacement of the protrusion **14*a*1** or the engagement groove **212**.

[0137] In the present embodiment, the two attachment holes **12** are separated in the x-axis direction, and the first attachment hole **12*a*** has a rectangular shape that is horizontally long in the x-axis direction as viewed in the z-axis direction. The distance between the two shaft members **210** may vary with respect to the distance between the two attachment holes **12** in the electronic component holder **100**. When the plurality of attachment holes **12** is provided in the electronic component holder **100**, the electronic component holder **100** can be attached to the attachment destination member even if the distance between the shaft members **210** varies by making one of the attachment holes **12** long in the separation direction in the separation direction between the attachment holes **12**.

[0138] Note that the present invention is not limited to the above-described embodiments, and includes various modifications, improvements, and the like as long as the object of the present invention is achieved.

[0139] The following modifications can be appropriately combined.

[0140] Although the electronic component holder **100** in the present embodiment has the two attachment holes **12**, the electronic component holder **100** may have only one attachment hole **12**, or may have three or more attachment holes **12**.

[0141] The leaf spring portion **11** may be provided at the peripheral edge portion of the flange portion **10**. That is, a part of the peripheral wall **14** defining the attachment hole **12** may be close to and along the peripheral edge portion of the flange portion **10**, and the peripheral edge portion of the flange portion **10** sandwiched between the peripheral edge of the flange portion **10** and a part of the peripheral wall **14** may be narrow to function as the leaf spring portion **11**.

[0142] In the present embodiment, the protrusion **14*a*1** is provided only in a part of the peripheral wall **14** of the attachment hole **12**, but is not limited thereto. The protrusion **14*a*1** may be formed over the entire circumference of the peripheral wall **14** of the attachment hole **12**.

[0143] In the first attachment hole **12*a***, the leaf spring portion **11** may be further provided on the facing wall portion **15** facing the leaf spring portion **11**. That is, the width dimension of the facing wall portion **15** may be reduced to such an extent as to have flexibility.

[0144] In the present embodiment, the leaf spring portion **11** is erected orthogonal to the lateral direction, that is, parallel to the longitudinal direction, but is not limited thereto. The leaf spring portion **11** may be inclined obliquely with respect to the longitudinal direction. Specifically, the leaf spring portion **11** may be inclined so as to fall toward the inside of the attachment hole **12**.

[0145] In the attached state, a nut may be further fastened to the shaft member **210** (bolt), and the nut and the installation surface **200** may be brought into pressure contact with the flange portion **10**. In this case, displacement of the flange portion **10** in the z-axis direction is prevented by the nut and the installation surface **200**. As described above, the flange portion **10** is also prevented from being displaced in the lateral direction by being gripped by the protrusion **14*a*1** and the facing portion **14*b***. In a case where a general nut is tightened to a bolt and the flange portion **10** is gripped by the installation surface **200** and the nut, and in a case where the flange portion **10** is gripped by the screw head and the installation surface **200** as in JP 2016-100610 A, the flange portion **10** is deformed due to aging, and the force of gripping the flange portion **10** by the installation surface **200** and the nut (or the screw head) is weakened. As in the present embodiment, by gripping the shaft member **210** in the lateral direction by the flange portion **10**, even if the gripping force of the flange portion **10** in the z-axis direction decreases, the relative positional relationship between the shaft member **210** and the electronic component holder **100** can be continuously maintained, and the electronic component holder **100** can be continuously held by the attachment destination member.

[0146] The above embodiment includes the following technical ideas.

[0147] (1) A vehicle electronic component holder set comprising: a shaft member having a circular cross section; and a vehicle electronic component holder, wherein

- **[0148]** the vehicle electronic component holder includes a base portion provided with an attachment hole through which the shaft member is inserted,
- **[0149]** the base portion includes a protrusion protruding from a peripheral wall defining the attachment hole toward an inside of the attachment hole,
- **[0150]** a distance between the protrusion and a center of the attachment hole is smaller than a radius of the shaft member as viewed from a depth direction of the attachment hole, and
- **[0151]** when the shaft member is inserted into the attachment hole, the protrusion comes into pressure contact with a side surface of the shaft member.

[0152] (2) The vehicle electronic component holder set according to (1), wherein

- **[0153]** a part of the base portion is a leaf spring portion extending along a periphery of the attachment hole,
- **[0154]** a pressed surface of the leaf spring portion is a part of the peripheral wall,
- **[0155]** each of both ends of the leaf spring portion in an extending direction of the leaf spring portion is formed continuously with an adjacent portion of the base portion adjacent to the leaf spring portion,
- **[0156]** a central portion of the leaf spring portion in the extending direction is separated from another part of the base portion, and
- **[0157]** the leaf spring portion is bendable so that an intermediate portion of the leaf spring portion is displaced toward the center of the attachment hole or toward a direction opposite to the center of the attachment hole.

[0158] (3) The vehicle electronic component holder set according to (2), wherein a dimension of the leaf spring portion in a direction orthogonal to the pressed surface that is a part of the peripheral wall is smaller than a dimension of the leaf spring portion in the depth direction of the attachment hole.

[0159] (4) The vehicle electronic component holder set according to (2) or (3), wherein

- **[0160]** a facing portion that is a part of the peripheral wall faces the pressed surface of the leaf spring portion across the attachment hole,
- **[0161]** at least a part of the pressed surface of the leaf spring portion is a forming region of the protrusion, and
- **[0162]** the facing portion is a non-forming region of the protrusion.

[0163] (5) The vehicle electronic component holder set according to any one of (2) to (4), wherein

- **[0164]** a facing portion that is a part of the peripheral wall faces the pressed surface of the leaf spring portion across the attachment hole,
- **[0165]** the base portion includes a facing wall portion that includes the facing portion and is located on a side away from the center of the attachment hole, and
- **[0166]** a dimension of the facing wall portion in a direction orthogonal to the facing portion is larger than a dimension of the leaf spring portion in a direction orthogonal to the pressed surface.

[0167] (6) The vehicle electronic component holder set according to any one of (2) to (5), wherein

- **[0168]** the base portion has a through hole provided in a vicinity of the attachment hole along a part of the peripheral wall, and
- **[0169]** the leaf spring portion is disposed between the attachment hole and the through hole.

[0170] (7) The vehicle electronic component holder set according to (6), wherein

- **[0171]** the through hole is elongated in the extending direction of the leaf spring portion, and
- **[0172]** a width of the through hole is smaller than a dimension of the leaf spring portion in a direction orthogonal to the pressed surface.

[0173] (8) The vehicle electronic component holder set according to any one of (2) to (7), wherein a top portion of the protrusion that protrudes most toward the inside of the attachment hole is a flat top surface extending in the depth direction of the attachment hole.

[0174] (9) The vehicle electronic component holder set according to (8), wherein

[0175] the side surface of the shaft member is provided with a bottomed groove portion that engages with the protrusion when the shaft member is inserted into the attachment hole, and

[0176] a dimension of the top surface in the depth direction of the attachment hole is larger than a width of a bottom portion of the groove portion in an axial direction of the shaft member.

[0177] (10) The vehicle electronic component holder set according to (9), wherein

[0178] the groove portion extends in a direction intersecting the axial direction of the shaft member,

[0179] at least one wall portion of two wall portions forming the groove portion so as to sandwich the groove portion in the axial direction is inclined with respect to the axial direction of the shaft member,

[0180] the protrusion has an inclined surface that is inclined with respect to the depth direction and faces the one wall portion when the shaft member is inserted into the attachment hole, and

[0181] an inclination of the inclined surface in the protrusion with respect to the depth direction of the attachment hole is smaller than an inclination of the one wall portion of the groove portion with respect to the axial direction.

[0182] (11) The vehicle electronic component holder set according to (10), wherein

[0183] the protrusion has an end surface continuing to a side opposite to the inclined surface with respect to the top portion,

[0184] the end surface faces a direction different from the inclined surface in the depth direction, and

[0185] an inclination of the end surface of the protrusion with respect to the depth direction is larger than the inclination of the inclined surface of the protrusion with respect to the depth direction.

[0186] (12) The vehicle electronic component holder set according to any one of (2) to (11), wherein

[0187] the attachment hole has a polygonal shape, and

[0188] the leaf spring portion is provided along a part of the peripheral wall forming each of at least two adjacent sides of the attachment hole.

[0189] (13) The vehicle electronic component holder set according to any one of (2) to (11), wherein

[0190] the attachment hole has a polygonal shape,

[0191] a length of one side of the attachment hole is larger than a length of another side of the attachment hole, and

[0192] the leaf spring portion is formed to include the one side.

[0193] (14) The vehicle electronic component holder set according to any one of (1) to (13), wherein

[0194] the vehicle electronic component holder includes a case portion that encloses an electronic component,

[0195] the case portion includes a bottomed recess that has an opening on a surface of the case portion and into which the electronic component is inserted,

[0196] the opening is disposed on one side of the base portion in the depth direction of the attachment hole,

[0197] the attachment hole is disposed outward of the case portion in a direction from a bottom portion of the recess toward the opening, and

[0198] an outer surface of the protrusion facing the one side is flush with a main surface of the base portion facing the one side.

[0199] (15) A vehicle electronic component holder used together with a shaft member having a circular cross section, the vehicle electronic component holder comprising:

[0200] a base portion provided with an attachment hole through which the shaft member is inserted, wherein

[0201] the base portion includes a protrusion protruding from a peripheral wall defining the attachment hole toward an inside of the attachment hole.

What is claimed is:

1. A vehicle electronic component holder set comprising: a shaft member having a circular cross section;

and a vehicle electronic component holder, wherein the vehicle electronic component holder includes a base portion provided with an attachment hole through which the shaft member is inserted,

the base portion includes a protrusion protruding from a peripheral wall defining the attachment hole toward an inside of the attachment hole,

a distance between the protrusion and a center of the attachment hole is smaller than a radius of the shaft member as viewed from a depth direction of the attachment hole, and

when the shaft member is inserted into the attachment hole, the protrusion comes into pressure contact with a side surface of the shaft member.

2. The vehicle electronic component holder set according to claim **1**, wherein

a part of the base portion is a leaf spring portion extending along a periphery of the attachment hole,

a pressed surface of the leaf spring portion is a part of the peripheral wall,

each of both ends of the leaf spring portion in an extending direction of the leaf spring portion is formed continuously with an adjacent portion of the base portion adjacent to the leaf spring portion,

a central portion of the leaf spring portion in the extending direction is separated from another part of the base portion, and

the leaf spring portion is bendable so that an intermediate portion of the leaf spring portion is displaced toward the center of the attachment hole or toward a direction opposite to the center of the attachment hole.

3. The vehicle electronic component holder set according to claim **2**, wherein a dimension of the leaf spring portion in a direction orthogonal to the pressed surface that is a part of the peripheral wall is smaller than a dimension of the leaf spring portion in the depth direction of the attachment hole.

4. The vehicle electronic component holder set according to claim **2**, wherein

a facing portion that is a part of the peripheral wall faces the pressed surface of the leaf spring portion across the attachment hole,

at least a part of the pressed surface of the leaf spring portion is a forming region of the protrusion, and

the facing portion is a non-forming region of the protrusion.

5. The vehicle electronic component holder set according to claim **2**, wherein a facing portion that is a part of the peripheral wall faces the pressed surface of the leaf spring portion across the attachment hole,

the base portion includes a facing wall portion that includes the facing portion and is located on a side away from the center of the attachment hole, and

a dimension of the facing wall portion in a direction orthogonal to the facing portion is larger than a dimension of the leaf spring portion in a direction orthogonal to the pressed surface.

6. The vehicle electronic component holder set according to claim **2**, wherein

the base portion has a through hole provided in a vicinity of the attachment hole along a part of the peripheral wall, and

the leaf spring portion is disposed between the attachment hole and the through hole.

7. The vehicle electronic component holder set according to claim **6**, wherein

the through hole is elongated in the extending direction of the leaf spring portion, and

a width of the through hole is smaller than a dimension of the leaf spring portion in a direction orthogonal to the pressed surface.

8. The vehicle electronic component holder set according to claim **2**, wherein a top portion of the protrusion that protrudes most toward the inside of the attachment hole is a flat top surface extending in the depth direction of the attachment hole.

9. The vehicle electronic component holder set according to claim **8**, wherein

the side surface of the shaft member is provided with a bottomed groove portion that engages with the protrusion when the shaft member is inserted into the attachment hole, and

a dimension of the top surface in the depth direction of the attachment hole is larger than a width of a bottom portion of the groove portion in an axial direction of the shaft member.

10. The vehicle electronic component holder set according to claim **9**, wherein

the groove portion extends in a direction intersecting the axial direction of the shaft member,

at least one wall portion of two wall portions forming the groove portion so as to sandwich the groove portion in the axial direction is inclined with respect to the axial direction of the shaft member,

the protrusion has an inclined surface that is inclined with respect to the depth direction and faces the one wall portion when the shaft member is inserted into the attachment hole, and

an inclination of the inclined surface in the protrusion with respect to the depth direction of the attachment hole is smaller than an inclination of the one wall portion of the groove portion with respect to the axial direction.

11. The vehicle electronic component holder set according to claim **10**, wherein

the protrusion has an end surface continuing to a side opposite to the inclined surface with respect to the top portion,

the end surface faces a direction different from the inclined surface in the depth direction, and

an inclination of the end surface of the protrusion with respect to the depth direction is larger than the inclination of the inclined surface of the protrusion with respect to the depth direction.

12. The vehicle electronic component holder set according to claim **2**, wherein

the attachment hole has a polygonal shape, and

the leaf spring portion is provided along a part of the peripheral wall forming each of at least two adjacent sides of the attachment hole.

13. The vehicle electronic component holder set according to claim **2**, wherein

the attachment hole has a polygonal shape,

a length of one side of the attachment hole is larger than a length of another side of the attachment hole, and

the leaf spring portion is formed to include the one side.

14. The vehicle electronic component holder set according to claim **1**, wherein

the vehicle electronic component holder includes a case portion that encloses an electronic component,

the case portion includes a bottomed recess that has an opening on a surface of the case portion and into which the electronic component is inserted,

the opening is disposed on one side of the base portion in the depth direction of the attachment hole,

the attachment hole is disposed outward of the case portion in a direction from a bottom portion of the recess toward the opening, and

an outer surface of the protrusion facing the one side is flush with a main surface of the base portion facing the one side.

15. A vehicle electronic component holder used together with a shaft member having a circular cross section, the vehicle electronic component holder comprising:

a base portion provided with an attachment hole through which the shaft member is inserted, wherein

the base portion includes a protrusion protruding from a peripheral wall defining the attachment hole toward an inside of the attachment hole.

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